

Unit 10 Overview

Arcs and Angle Relationships in Circles



In this unit, students will extend their knowledge of circles to include new vocabulary and relationships such as sectors, arcs, radians, central angles, inscribed angles, chords, secants, and tangents. Most of the content in this unit will be new vocabulary and concepts to students, regardless of their mathematical progression prior to taking Geometry.

Section 1 provides students the opportunity to work with arc lengths and sectors. Students will learn the concept of a sector and its relationship to the circle. Using the circumference of the circle, they will learn the concept of an arc and how to find the length of an arc. In Lesson 3, students will use what they have learned about arc length to the radius to learn about the definition of a radian and how to measure angles in radian units. Lesson 4 is an opportunity for students to bring together everything that they have learned in Section 1 to solve problems involving algebraic expressions with arc length and area of a sector.

In Section 2, students focus primarily on arc measures and related angles. Students will learn about the relationship between the measure of a central angle and its intercepted arc, using this relationship to solve problems. Students then move from central angles to inscribed angles. Students are introduced to intersecting chords, secants, and tangents, and the relationships that exist between the angles formed and the intercepted arcs. For students enrolled in Geometry Honors, Lesson H1 focuses on constructing a line tangent to a circle that passes through a given point outside the circle.

Unit At-A-Glance

Section 1: Arc Lengths and Sectors	Benchmarks Addressed	Lesson 1	Sectors
	MA.912.GR.6.4	Lesson 2	Arc Length
		Lesson 3	Radian Measure
		Lesson 4	Solving Problems Involving Arc Length and Area of a Sector
Section 2: Arc Measures and Related Angles	Benchmarks Addressed	Lesson 5	Solving Problems Involving Arc Measure and Central Angles
	MA.912.GR.6.2 MA.912.GR.5.5 (<i>Honors only</i>)	Lesson 6	Solving Problems Involving Inscribed Angles
		Lesson 7	Solving Problems Involving Intersecting Chords
		Lesson 8	Solving Problems Involving Intersection of Secants
		Lesson 9	Solving Problems Involving Tangents
		Lesson H1	Constructing Tangent Lines



Differentiation

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Engagement <i>The “WHY” of learning</i>	- To increase student engagement, consider making parts of this lesson digital. For Exploration 10.6.2, consider supporting the activity through the use of applets like GeoGebra’s Central Angles, Inscribed Angles, and Intercepted Arcs activities. - In the components where students are asked to work independently, consider allowing students who struggle to pair up with a partner. This will ensure students are engaged in a productive struggle and sustain effort. - The Wrap-Ups in this unit provide students with multiple opportunities to self assess and reflect.
Representation <i>The “WHAT” of learning</i>	- Consider creating anchor charts when students bring together their explorations and the formal introductions to vocabulary and angle relationships to arcs. Anchor charts will assist students with recalling the relationships based on the different components. - Some learners may benefit from seeing how angles are formed by secants to better understand this conceptually. To do this, consider GeoGebra’s Angles formed by Secants (outside) digital display. - Auditory entry points are available to students in the Collaboration activities throughout the unit. Partner discussion will help students gain confidence with and understanding of the learning targets.
Action & Expression <i>The “HOW” of learning</i>	- Several components include constructive-response problems that require students to explain why or how they solved the problem. The teacher guides include alternative methods for students who struggle expressing themselves in writing, such as using an audio recorder or speech-to-text software. - In the Wrap-Ups throughout the unit, allow students to demonstrate their mastery of the learning targets using different methods, such as writing a reflective self-assessment, solving example problems, or drawing a visual model. 1. I can define a chord of a circle. I need help, I can't get started. I'm getting there, but I need more practice. I understand this, and I feel confident. 2. I can define the relationship of angles and arcs formed by the intersection of two chords. I need help, I can't get started. I'm getting there, but I need more practice. I understand this, and I feel confident. 3. I can solve problems involving chords of a circle. I need help, I can't get started. I'm getting there, but I need more practice. I understand this, and I feel confident.

ELL Information

Focus on vocabulary and language development, guided interaction, authentic assessment, explicit instruction, meaning-based context and universal themes, modeling, graphic organizers, visuals, and group work.

- This unit is vocabulary intensive. For students whose first language is not English, vocabulary assistance should be offered for keywords throughout the topic. Encourage students to use a dual language dictionary or technology to translate or look up the meaning of the words.
- Prominently display a visual aid of a large circle, and use it as a reference throughout the unit. Add the new vocabulary words to the diagram, illustrating them with new colors. Use coordinating colors to add the formula for relationships between angles and arcs.
- Visually connect the measure of the arc to the measure of its central angle; highlight that the measure of the arc, x , is a fraction of the entire circle.
- In word problems and in diagrams, have students highlight important vocabulary words, such as center, radius, diameter, arc, arc length, etc.
- Allow students to work in partners or triads, when appropriate; carefully plan small groups so that ELL students are partnered with non-ELL students.
- Provide frequent opportunities for students to turn-and-talk; pair up ELL students with non-ELL students.
- Encourage students to make dual-language flashcards of the vocabulary words, properties, and theorems, including diagrams where appropriate.
- Provide examples for an interactive notebook.

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support” below:

- Central Angle
- Circumference
- Diameter
- Equilateral triangle
- Inscribed angle in a circle
- Proportional relationship
- Radius

Additional Differentiated Support

Scaffolding Support	On-Ramp
• Play a memory or matching game with definitions. • Example Activity: Teach the mnemonic for circumference formula: “Cherry Pie Is Delicious.” See what other clever mnemonics the students can create! • Example Activity: Explore circumference and diameter by measuring circular objects. You will need a ruler, string, and flexible tape measurer. To measure the circumference, students can use a string and then measure the length of the string, or use a flexible tape measurer. • Example Activity: Create large circles of different sizes on the floor with tape. Have the students walk around the circle and then determine how far they have walked. Provide students with yardsticks, tape measurers, and rulers. Encourage them to first identify the diameter.	Use the videos and practice problems in the On-Ramp to Geometry personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish. • Area and Circumference of a Circle



Section Overview

Section 1: Arc Length and Sectors (Lessons 1-4)

Benchmarks Addressed	Learning Targets	
MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle. <i>Clarification 1: Instruction focuses on the conceptual understanding that the length of the arc intercepted by an angle is proportional to the radius.</i>	Lesson 1	- I can define a sector of a circle. - I can explain how the area of a sector is defined as a fractional part of the area of a circle.
	Lesson 2	- I can define “arc length.” - I can explain how the length of the arc is proportional to the ratio of the central angle to the whole circle, measured in degrees.
	Lesson 3	- I can define “radian.” - I can explain the relationship between the length of the arc intercepted by an angle to the radius.
	Lesson 4	I can solve mathematical and real-world problems involving arc length and sector area.

Section 2: Arc Measures and Related Angles (Lessons 5-9 & H)

Benchmarks Addressed	Learning Targets	
MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles. <i>Clarification 1: Within the Geometry course, problems are limited to relationships between inscribed angles; central angles; and angles formed by the following intersections: a tangent and a secant through the center, two tangents, and a chord and its perpendicular bisector.</i> MA.912.GR.5.5 (Honors only) Given a point outside a circle, construct a line tangent to the circle that passes through the given point. <i>Clarification 1: When given a circle, the center must be provided.</i> <i>Clarification 2: Instruction includes using a compass and straightedge, string, reflective devices, paper folding or dynamic geometric software.</i>	Lesson 5	I can use the relationship between the measure of a central angle and its intercepted arc to solve problems.
	Lesson 6	I can use the relationship between the measure of an inscribed angle and its intercepted arc to solve problems.
	Lesson 7	I can solve problems involving chords of a circle and their related angles.
	Lesson 8	I can solve problems involving secants drawn from a point external to the circle and their related angles.
	Lesson 9	I can solve problems related to tangents and secants, and the measures of their intercepted arcs.
	Lesson H1	I can construct a line tangent to a circle that passes through a given point outside the circle.



Vertical Alignment

Geometry Unit 10
Arcs and Angle Relationships in Circles

Course Level

MA.912.GR.6.4

Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.

MA.912.GR.6.2

Solve mathematical and real-world problems involving the measures of arcs and related angles.

MA.912.GR.5.5 (*Honors only*)

Given a point outside a circle, construct a line tangent to the circle that passes through the given point.

10.1 Sectors

Lesson Introduction

 Benchmark	MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.		 Learning Targets	- I can define a sector of a circle. - I can explain how the area of a sector is defined as a fractional part of the area of a circle.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- What is the relationship between the degree measure of a central angle and the degree measure of an entire circle? - How can you calculate the area of a sector when given a circle using ratios?			
 Lesson Narrative	Students will explore central angles and their measures in relation to the measure of the entire circle. Using this relationship, students will be able to find the area of a sector by using proportions.			
 Component Summary	10.1.1 Warm-Up (~5 min.)	Speculating about the game of Pac-Man (<i>SE p. 119</i>)	10.1.3 Guided Practice (10-15 min.)	Finding the area of sectors using proportions (<i>SE p. 121</i>)
	10.1.2 Exploration (10-15 min.)	Exploring fractions of a circle (<i>SE p. 120</i>)	10.1.4 Your Turn! (5-10 min.)	Calculating fractions, areas, and central angles of the sectors of a circle (<i>SE p. 123</i>)
	10.1.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 124</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Central Angle <u>Additional Glossary:</u> - Sector of a Circle		Video for Warm-Up 10.1.1 (optional) Calculator	Warm-Up 10.1.1 contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions		Things to Consider
	- Students may have difficulty manipulating the expressions to isolate the sector area in the 10.1.3 Guided Practice. - Students may use the formula of the circumference to calculate the area of a circle. - Students may struggle with simplifying the ratio of the central angle to the whole circle, and not see the connection between the ratio and the sector fraction of the circle.		- Facilitate a conversation about the difference between finding the area of a circle and finding the circumference of a circle. - Students should not be using the formal formula for calculating the area of a sector yet.
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard
	Clarify, Critique, Connect Consider implementing this strategy by allowing students to trade their response in the 10.1.3 Guided Practice or the 10.1.4 Your Turn! with another student. Students will then ask their partner clarifying questions as needed. This process will allow students to think more analytically and provide a more detailed mathematical explanation.		MA.K12.MTR.1.1: Participate in Effortful Learning Students will be working on finding the area of a sector by making connections between the fraction of a circle and the central angle. As students explore this connection, they need to ask clarifying questions to deepen their understanding. The 10.1.2 Exploration and the 10.1.3 Guided Practice provide help and support for students with the new concept. The 10.1.4 Your Turn! will provide students the confidence that they need to complete and work through complex and/or new problems.
 Differentiate Instruction	This lesson provides multiple visual entry points for students as they discover the relationship between the measure of the central angles, the area of the sector, and the area of the circle. Read and discuss problems aloud to assist students who benefit from auditory entry points.		
Lesson Notes:			

10.1.1 Warm-Up: Pac-Man

Component Overview: During this Warm-Up, students will watch a short video (33 seconds) of Pac-Man going around the maze and eating dots. Students will make observations by writing down things they notice and wonder about Pac-Man's body as he opens and closes his mouth to eat dots.



10.1.1 Warm-Up: Pac-Man

The classic and popular Pac-Man video game was released in Japan in May of 1980 and was in the United States by October of that same year. The yellow, pie-shaped character travels around a maze trying to eat dots while avoiding the ghosts hunting him.

Watch the video of Pac-Man going around the maze and eating dots.



What do you notice about the location of Pac-Man's mouth in relation to his body? How is Pac-Man's body with an open mouth similar to a pizza missing one slice?

1. What do you notice about Pac-Man's body as he opens and closes his mouth to eat the dots?

Answers will vary. I noticed that his body is a whole circle when his mouth is closed and is a partial circle when his mouth is open. I noticed that his body get smaller when his mouth is opening and larger when it's closing.



2. What do you wonder as you watch Pac-Man move around the maze?

Answers will vary. I wonder why he doesn't get larger as he consumes each pellet. I wonder what would cause Pac-Man to go faster throughout the maze. I wonder how many power pellets can fit inside Pac-Man and I wonder what those creatures are chasing him.

10.1.2 Exploration: Fractions of a Circle

Component Overview: Students will be calculating the area of a given number of pizza slices. Then, students will be exploring the relationship between the fraction of a circle and its central angle in degrees. Students will have the opportunity to work with a partner to create a definition for a sector of a circle as it relates to a slice of pizza.



10.1.2 Exploration: Fractions of a Circle

Work with your partner to complete the following.

1. Pizza sizes are given in terms of the pizza's diameter. The standard size for most large circular pizzas is 14 inches (in.) and it is usually cut into 10 triangular slices.



- a. What is the approximate area of a 14-inch pizza? Use $\frac{22}{7}$ for π .

$$A = \pi r^2$$

$$r = \frac{\text{diameter}}{2} = \frac{14}{2} = 7$$

$$A = \left(\frac{22}{7}\right)(7)^2$$

$$A = \left(\frac{22}{7}\right)49$$

$$A \approx 154 \text{ in}^2$$

- b. If the slices are cut precisely and equally, what fraction and percentage of the pizza is 5 slices?

Fraction	Percentage
$\frac{5}{10} = \frac{1}{2}$	50%

- c. What is the area of 5 slices?

$$\frac{154 \text{ in}^2}{2} \approx 77 \text{ in}^2$$

- d. What is the area of 1 slice?

$$\frac{154 \text{ in}^2}{10} \approx 15.4 \text{ in}^2$$



Consider prompting questions to help students make sense of the problem as a check for understanding.

- What is the radius of the circle?
- How can you find the area of a circle?
- What does it mean to use $\frac{22}{7}$ for π ?



Consider challenging students to find the area of 2 slices of pizza if the 14-inch pizza was cut into 8 triangular slices.

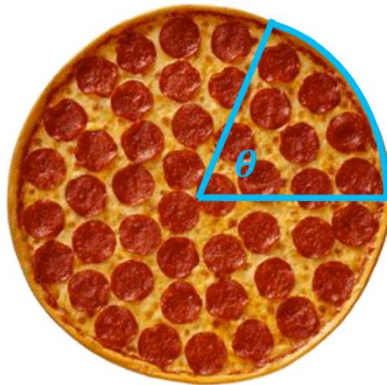
Have students use $\frac{22}{7}$ for π . (38.5 in^2)

10.1.2 Exploration: Fractions of a Circle (continued)

2. The point where a pizza is divided to create precisely equal slices is a central angle.



A **central angle** is an angle with its vertex at the center of a circle with radii as its side.



The central angle is a fraction of the total 360° in a circle.

- a. Complete the table. The first row of the table has been filled in.

	Fraction	Central Angle in Degrees ($^\circ$)
Half of a Circle	$\frac{1}{2}$	$\left(\frac{1}{2}\right) \cdot (360^\circ) = 180^\circ$
Quarter of a Circle	$\frac{1}{4}$	$\left(\frac{1}{4}\right) \cdot (360^\circ) = 90^\circ$
Third of a Circle	$\frac{1}{3}$	$\left(\frac{1}{3}\right) \cdot (360^\circ) = 120^\circ$
Tenth of a Circle	$\frac{1}{10}$	$\left(\frac{1}{10}\right) \cdot (360^\circ) = 36^\circ$



Notice and Wonder

Consider providing students with the opportunity to write down things they notice and wonder about the central angle. This will allow students the chance to make observations to deepen their understanding of a central angle.



What relationship do you notice between the fraction of the circle and the central angle in degrees? How is this similar to finding the area of a given number of slices of pizza?

10.1.2 Exploration: Fractions of a Circle (continued)

- b. Complete the statement.

One slice, or fraction of the circle, is bounded by two
☐ diameters
☒ radii and an arc. Since each slice is the same
 size, the central angle for each fractional piece of the
 circular pizza will ☐ vary.
☒ be equal.

- c. The pizza slices are an example of **sectors**. With your partner, write a definition of the term **sector**.

Answers may vary.

A sector is a fraction of the area of a circle.

A sector is a piece of a circle that is in the shape of a slice of pizza, with a maximum central angle of 360° .

- d. Ask two other partner pairs to share their definition of a sector. Record a summary of their definition. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was correct.

My Summary of Their Definition of a Sector	Initials
<i>Answers will vary. A sector is a fraction of a circle with a central angle.</i>	
<i>Answers will vary. A sector is a piece of a circle, that looks like a slice of pizza, that represents a fraction of the circles' area.</i>	



For students who may struggle with showing their understanding of a sector, consider alternative ways to assess their learning and understanding such as recording themselves, or drawing and labeling a picture.

10.1.3 Guided Practice: Finding the Area of a Sector

Component Overview: Students will build a connection between the sector fraction of a circle and the central angle of the sector. Students will have the opportunity to find the area of a given sector using the relationship between the sector fraction of a circle and the central angle of a sector.



10.1.3 Guided Practice: Finding the Area of a Sector

The images below show a stopwatch with different time increments marked. The blue shading on the stopwatches each forms a sector.

1. Complete the table.

	Sector Fraction of the Circle	Central Angle Measure of the Sector	Ratio of the Central Angle to the Whole Circle
	$\frac{1}{12}$	$\left(\frac{1}{12}\right) \cdot (360^\circ) = 30^\circ$	$\frac{30}{360} = \frac{1}{12}$
	$\frac{9}{12}$	$\left(\frac{9}{12}\right) \cdot (360^\circ) = 270^\circ$	$\frac{270}{360} = \frac{3}{4}$
	$\frac{2}{12}$	$\left(\frac{2}{12}\right) \cdot (360^\circ) = 60^\circ$	$\frac{60}{360} = \frac{1}{6}$

2. Summarize what you notice about the relationship between the values in each row.

Answers may vary. The sector fraction of a circle is equal to the ratio of the central angle to the whole circle.



Think-Pair-Share

This strategy can be used on questions 1 and 2.

- Students should work each problem individually and record answers.
- Students will pair with their partner or group members to discuss solutions and provide constructive critiques.
- The teacher will select a few groups to share their solutions and methods.



Discussion provides additional entry points for auditory learners. Consider utilizing sentence stems, and provide ample time for students to grapple with academic language.

- I think the central angle is _____ because the fraction of the circle is _____.
- I noticed that the sector fraction and the ratio of the central angle to the whole circle are _____ which makes me think _____.

10.1.3 Guided Practice: Finding the Area of a Sector (continued)

3. Discuss with your partner how ratios could be used to find the area of the sectors, including any additional information that might be needed. Summarize your discussion.

Answers may vary.

$$\frac{\text{Central Angle}}{360^\circ} = \frac{\text{Area of Sector}}{\text{Area of Circle}}$$

$$\text{So, Area of Sector} = \frac{\text{Area of Circle} \cdot \text{Central Angle}}{360^\circ}$$

In order to solve for the area of a sector, the area of the circle must be known so, additional information needed would be that of the radius.

Given the radius of the stopwatch is 2 in., the area of the stopwatch is 4π in.².

$$\begin{aligned} A &= \pi r^2 \\ A &= \pi(2)^2 \\ A &= 4\pi \text{ in.}^2 \end{aligned}$$



What does the area of a sector represent? How can proportions help you to find the area of a sector? What information will you need to find the area of a sector?

10.1.3 Guided Practice: Finding the Area of a Sector (continued)



10.1.3 Guided Practice: Finding the Area of a Sector

The images below show a stopwatch with different time increments marked. The blue shading on the stopwatches each forms a sector.

- Complete the table.

	Sector Fraction of the Circle	Central Angle Measure of the Sector	Ratio of the Central Angle to the Whole Circle
	$\frac{1}{12}$	$\left(\frac{1}{12}\right) \cdot (360^\circ) = 30^\circ$	$\frac{30}{360} = \frac{1}{12}$
	$\frac{9}{12}$	$\left(\frac{9}{12}\right) \cdot (360^\circ) = 270^\circ$	$\frac{270}{360} = \frac{3}{4}$
	$\frac{2}{12}$	$\left(\frac{2}{12}\right) \cdot (360^\circ) = 60^\circ$	$\frac{60}{360} = \frac{1}{6}$

- Summarize what you notice about the relationship between the values in each row.

Answers may vary. The sector fraction of a circle is equal to the ratio of the central angle to the whole circle.



Emphasize that the sector area is a fraction of the area of the circle. Students should not be using the formal formula for calculating area of a sector yet.

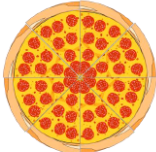
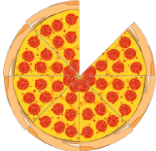
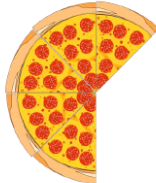
10.1.4 Your Turn!

Component Overview: Students will work individually to find the fraction of the circle, the area of the sector, and the central angle. Students will leave their answer for the area of the sector in terms of π .



10.1.4 Your Turn!

- Kenya orders a medium pizza that has a 8 in. radius. Complete the table on the number of slices she has.

	Fraction of the Circle	Area of the Sector	Central Angle
Eight Slices 	$\frac{8}{8} = 1$	$\pi(8)^2$ $64\pi \text{ in}^2$	$1 \cdot 360^\circ$ $= 360^\circ$
Seven Slices 	$\frac{7}{8}$	$\frac{7}{8} \cdot 64\pi$ $= 56\pi \text{ in}^2$	$\frac{7}{8} \cdot 360^\circ$ $= 315^\circ$
Five Slices 	$\frac{5}{8}$	$\frac{5}{8} \cdot 64\pi$ $= 40\pi \text{ in}^2$	$\frac{5}{8} \cdot 360^\circ$ $= 225^\circ$
Three Slices 	$\frac{3}{8}$	$\frac{3}{8} \cdot 64\pi$ $= 24\pi \text{ in}^2$	$\frac{3}{8} \cdot 360^\circ$ $= 135^\circ$

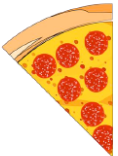


Emphasize to students what it means to leave an answer in terms of π .



Consider challenging students who completed their work early to create a formula that can be used to find the area of a sector of any circle. Encourage students to include the central angle in the formula.

10.1.4 Your Turn! (continued)

One Slice 	$\frac{1}{8}$	$\frac{1}{8} \cdot 64\pi$ $= 8\pi \text{ in}^2$	$\frac{1}{8} \cdot 360^\circ$ $= 45^\circ$
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Consider using this time to explore more about what the students may still have questions about. This can be done in small groups, or by checking in on students.

10.1.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



10.1.5 Wrap-Up: Cool Down

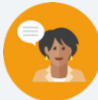
1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about sectors.

Answers will vary.

10.2 Arc Length

Lesson Introduction

 Benchmark	MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.		 Learning Targets	- I can define arc length. - I can explain how the length of the arc is proportional to the ratio of the central angle to the whole circle, measured in degrees.	
 Benchmark Coherence	Building On		Working Toward		
	N/A		N/A		
 Essential Questions	- What is an arc length of a circle? - How is the length of an arc proportional to the ratio of the central angle to the whole circle? - What is the relationship between the length of an arc and the circumference of a circle?				
 Lesson Narrative	In this lesson, students will learn that the length of an arc varies directly with the central angle, and is a fraction of the circumference of the entire circle. Students will discover that the ratio of the arc length to the circumference of the circle is equal to the ratio of the central angle to the whole circle.				
 Component Summary	10.2.1 Warm-Up (~5 min.)	Examining an image of the Chandelier Tree (<i>SE p. 125</i>)	10.2.3 Activity (10 min.)	Matching central angles and ratios of arc length to circumference (<i>SE p. 127</i>)	
	10.2.2 Exploration (15-20 min.)	Discovering the relationship between arc length and central angles (<i>SE p. 125</i>)	10.2.4 Exploration (10 min.)	Discovering the relationship of central angles between circles of different radius measures (<i>SE p. 128</i>)	
	10.2.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p.129</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> Arc Length		- Cards for Card Sort (optional) String		N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may have difficulty simplifying fractions to see the connection between the ratio of the arc length to the circumference, and the ratio of the central angle to the whole circle.	- Facilitate a conversation to distinguish between arc measure and arc length. - Students should not be using the formal formula for calculating the arc length yet. - If time is an issue, then consider assigning 10.2.4 Exploration for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect The lesson provides students with the opportunity to discover the relationship between the ratio of the central angle to the whole circle, and the ratio of the arc length to the circumference. Encourage students to compare and connect their findings to the discoveries from the previous lesson about sector area to the new discoveries made in this lesson. Allow students the chance to review examples from the lesson, so they can deepen their understanding about central angles.	MA.K12.MTR.4.1: Discussion and Reflection Use the 10.2.2 Exploration and the 10.2.3 Activity to encourage students to share their mathematical thinking with peers. Provide encouragement by creating a safe environment for students to communicate their mathematical ideas and accept feedback. Encourage students to use mathematical vocabulary during discussions.
 Differentiate Instruction	This lesson provides multiple entry points for visual, auditory, and kinesthetic learners. Students draw their own conclusions and discuss their rationale with peers and teachers, providing an opportunity for auditory learners to verbalize their conception. Consider providing string for kinesthetic learners to use for a hands-on interaction for learning about arc length. Images are shown throughout the lessons for visual learners to see the connection between the central angle and the arc length.	
Lesson Notes:		

10.2.1 Warm-Up: Notice and Wonder

Component Overview: Students will make observations about what they notice and wonder about the image. Students will estimate the diameter of the base of the tree and justify their answer.



10.2.1 Warm-Up: Notice and Wonder

The Chandelier Tree, located in Leggett California, is part of a drive-thru tree park. The tree is believed to have been carved in the early 1930's by Charlie Underwood.

1. What do you notice?

Answers will vary. I noticed that the trunk of the tree is very wide. I noticed that tree looks very sturdy. I notice that the tree is gigantic.



2. What do you wonder?

Answers will vary. I wonder how old the tree is. I wonder how they knew the tree could still stand after that much being carved out. I wonder how that hole got there.

3. Estimate the diameter of the tree at the base.

Answers will vary. I estimate the diameter of the tree to be 12 feet. I estimate the diameter of the tree to be 20 feet.

4. Explain why you chose that number as your estimate.

Answers will vary. I chose this number because it needs to be large enough for a car to go through. I chose this number because I estimated the length of the car in the photo (roughly 15 feet) and added a few more feet since the tree diameter is larger than the car.



Notice and Wonder

Utilize the built-in "Notice and Wonder" routine while students analyze the information and the image. Position students to share their findings with a partner to expand their curiosity.

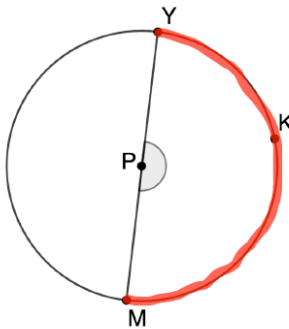
10.2.2 Exploration: Arc Length and Proportional Relationships

Component Overview: In this Exploration, students will build a connection between the central angle and the fraction of the circumference of a circle. Students will discover that the arc length is the distance along the circumference of a circle, and that the measure of the arc is proportional to the ratio of the central angle to the whole circle.



10.2.2 Exploration: Arc Length and Proportional Relationships

1. Circle P with point K on the circle and diameter $\overline{YM}$ is shown.



- a. What is $m\angle YPM$?
 180°
- b. Determine the ratio of the measure of the central angle, $\angle YPM$, to the whole circle. Simplify the ratio.
 $\frac{180^\circ}{360^\circ} = \frac{1}{2}$
- c. An arc of a circle is the portion of the circumference between points. Point K is a point on Circle P . Discuss with your partner which part of the circle is arc YKM . Then, trace over this portion of the circle.
Students do not need to write an explanation here, but a sample response is provided for reference.
Since arc YKM of the circle is 180° , arc YKM is half of the circle.
Tracing is on the image of Circle P above.
- d. What fraction of the circle's circumference does arc YKM represent?
 $\frac{1}{2}$



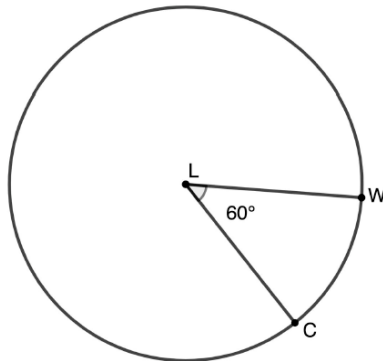
Consider prompting questions to help students make sense of the problems, as a check for understanding.

- What is a central angle of a circle?
- What is the circumference of a circle?

10.2.2 Exploration: Arc Length and Proportional Relationships (continued)

Arc YKM can also be notated as $\widehat{YKM}$.

2. Circle L is shown where points W and C are on the circle.



- a. What is the ratio of the central angle, $\angle CLW$, to the whole circle? Simplify the ratio.

$$\frac{60^\circ}{360^\circ} = \frac{1}{6}$$
- b. What fraction of the circle's circumference does $\widehat{WC}$ represent?

$$\frac{1}{6}$$
3. Verify your answers to problems 1 and 2 with your partner and resolve any differences, if necessary.
Answers will vary.
 $\frac{180}{360} = \frac{1}{2}$, so, any arc that is 180° , will be half the circumference of a circle. Any arc that is 60° , will be $\frac{1}{6}$ of the total circumference of a circle.



Notice and Wonder

Consider providing students with the opportunity to write down things they notice and wonder about the relationship between the ratio of the central angle to the whole circle, and the fraction of the circle's circumference the arc makes. This will allow students the chance to make observations to deepen their understanding of arc lengths before discussing with a partner.



Students do not need to write an explanation here, but a sample response is provided for reference.

10.2.2 Exploration: Arc Length and Proportional Relationships (continued)

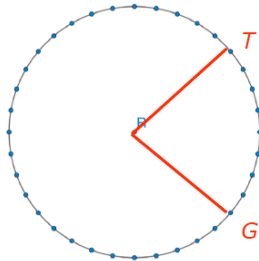
4. With your partner, discuss any relationships you observe. Summarize your discussion.

Answers may vary.

The ratio of the central angle to the whole circle is equivalent to the fraction of the circumference that the arc represents.

A fraction of a circle's circumference that an arc represents is the same as the ratio of the central angle to the whole circle.

5. On Circle R , the distance between each point measured along the circumference is 1 inch.



- a. Mark a point anywhere along the circle's circumference as point T . Then, starting at point T , mark an 8-inch section along the circle's circumference. Label the other endpoint of the 8-inch section as point G .

Answer on image above. Point locations will vary.

- b. What fraction of the circle's circumference does the length of $\widehat{TG}$ represent?

$$\frac{8\text{in}}{36\text{in}} = \frac{2}{9}$$

- c. Draw the central angle $\angle TRG$.

Answer on image above. Locations may vary.

- d. What is the measure of the central angle $\angle TRG$?

$$\frac{2}{9}(360^\circ) = 80^\circ$$



Consider giving students string to place on the circumference of the circle to create arc TG . This hands-on approach will help kinesthetic and visual learners visualize the arc in relation to the circumference of the circle.



Clarify, Critique, Connect

Consider providing different lengths of string to engage partners in a discussion about their findings. Students can trade their responses with their partner, then ask their partner clarifying questions, if needed. This process will allow students to think more analytically about the relationship between the arc length and the central angle.

10.2.2 Exploration: Arc Length and Proportional Relationships (continued)

- e. What is the ratio of the central angle to the whole circle?

$$\frac{80^\circ}{360^\circ} = \frac{8}{36} = \frac{2}{9}$$

- f. Discuss with your partner the relationship between the length of the arc and the ratio of the central angle to the whole circle.

Students do not need to write an explanation here, but a sample response is provided for reference.

The ratio of the length of the arc to the circumference of the circle is the same as the ratio of the central angle to the whole circle.

6. Complete the statements.

The ☐ arc length ☐ diameter ☐ radius is the distance along the part of

a circle's circumference or along any curve.

The measure of an arc is ☐ the inverse of ☐ proportional to the ratio of the central angle to the whole circle.



Consider asking the following to check for understanding before moving on:

- How is an arc length related to the circumference of a circle?
- What is the relationship between the measure of an arc and the ratio of the central angle to the whole circle?

10.2.3 Activity: Matching Central Angles and Ratios of Arc Length to Circumference

Component Overview: Students will work with their partner to match each circle to the correct ratio of the arc length to the circumference of the circle indicated on the card.



10.2.3 Activity: Matching Central Angles and Ratios of Arc Length to Circumference

1. Your teacher will give you two sets of cards for this activity. One set shows circles with a central angle. The other set of cards is a set of ratios of the arc length to the circumference.
 - a. Work with your partner to match each circle card to the corresponding ratio indicated on the lettered card. Record the matches in the table.

Card Letter	Ratio of the Arc Length to the Circumference	Card Letter	Ratio of the Arc Length to the Circumference
A	$\frac{1}{6}$	F	$\frac{5}{18}$
B	$\frac{1}{3}$	G	$\frac{2}{9}$
C	$\frac{7}{18}$	H	$\frac{1}{2}$
D	$\frac{1}{8}$	I	$\frac{5}{12}$
E	$\frac{1}{12}$	J	$\frac{5}{24}$

- b. Summarize how you and your partner were able to match the circles to a corresponding ratio without being given the circumference. *Answers may vary. The circles were matched to the corresponding ratios by finding the ratio of the central angle to the whole circle, which is equivalent to the ratio of the arc length to the circumference.*
 - c. Complete the statement.
The ratio of the central angle to the whole circle is the

☐ inverse of
☒ same as

 the ratio of the length of an arc to the circumference.



Consider asking the following while monitoring students during the activity to check for understanding:

- How is the central angle used to find the ratio of the arc length to the circumference?
- Will you always need the central angle to find the measure of an arc of a circle? Explain.



Consider challenging students who completed their work early to create a formula that can be used to find the measure of an arc of a circle. Encourage students to include the central angle in the formula.

10.2.4 Exploration: Arc Lengths With the Same Central Angle and Different Radii

Component Overview: Students will discover that when circles have the same central angle, the ratio of the arc length to the circumference will equal the ratio of the central angle to the whole circle. Students will justify their discovery by calculating the ratios for arc lengths to the circumference, and the ratio of the central angle to the whole circle.



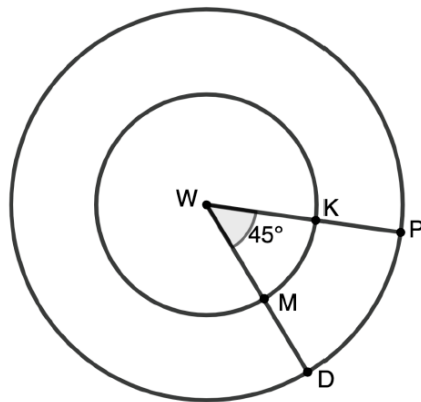
10.2.4 Exploration: Arc Lengths with the Same Central Angle and Different Radii

- Points D and P lie on Circle W . The radius $\overline{WP}$ measures 5 inches (in.).

A smaller Circle W with a radius measure of 2 in. rests inside the larger Circle W , where points M and K lie on the smaller circle.

The smaller Circle W has central angle $\angle KWM$ that measures 45° and the larger Circle W has central angle $\angle PWD$ that measures 45° .

Use the information to complete the following.



- $\widehat{MK}$ and $\widehat{DP}$ are different lengths. Discuss with your partner whether you think the ratio of the arc length and circumference will be the same as or different than the ratio of the central angle of the whole circle.

Students do not need to write an explanation here, but a sample response is provided for reference. We think the ratio will be the same because they have the same central angle.



Notice and Wonder

Consider providing students with the opportunity to write down things they notice and wonder about the image of smaller Circle W inside larger Circle W . This will provide students with the opportunity to analyze the image before discussing with a partner.

10.2.4 Exploration: Arc Lengths With the Same Central Angle and Different Radii (continued)

- b. Andrew and Sarah are discussing if the ratio of the arc length to the circumference is the same as the ratio of the central angle to the whole circle for both the smaller and larger circles.

Sarah says, "Yes. They will have the same ratios because they have the same central angle."

Andrew says, "No. They do not have the same ratios because they are different circles with different radii."

Explain who you agree with.

Answers will vary. Sarah is correct, the ratio of the central angle to the whole circle is the same as the ratio of the arc to the circumference.

- c. Complete the table to support your reasoning in Part B.

	Ratio of Arc Length to Circumference	Ratio of Central Angle to Whole Circle
Circle W with radius 2 in.	$\frac{1}{8}$	$\frac{45^\circ}{360^\circ} = \frac{1}{8}$
Circle W with radius 5 in.	$\frac{1}{8}$	$\frac{45^\circ}{360^\circ} = \frac{1}{8}$

- d. Explain whether your findings in Part C support your answers to Parts A and B. If not, explain why your answers to Parts A and B are incorrect.

Answers will vary. My findings support my answers to Parts A and B, this is because the ratios are the same.

My findings do not support my answers to Parts A and B. This is because the ratio of the arc length to the circumference were not the same as the ratio of the central angle to the whole circle.



To strengthen the conception of concentric circles, consider asking students:

- Will all circles with the same central angle have the ratio for arc length to circumference equal the ratio of the central angle to whole circle?

10.2.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



10.2.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

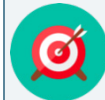
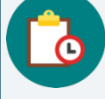
Answers will vary.



Consider collecting data from the cool down to gain insight on student understanding of the lesson. If many students indicate misconceptions about arc length, a review may be necessary prior to Lesson 3.

10.3 Radian Measure

Lesson Introduction

 Benchmark	MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.		 Learning Targets	- I can define radian. - I can explain the relationship between the length of the arc intercepted by an angle to the radius.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is a radian? - What are radians used for? - What is the relationship between the length of the arc intercepted by an angle and the radius of the circle?			
 Lesson Narrative	The focus of this lesson is for students to be able to define radian and explain the relationship between the radius, central angle, and arc length. Students will be looking at the circumference and arc length in circles of various radii to make discoveries about this relationship. They will be able to define radian as a ratio of arc length to radius.			
 Component Summary	10.3.1 Warm-Up (~5 min.)	Choosing between two bank deposit scenarios (<i>SE p. 130</i>)	10.3.3 Exploration (10-15 min.)	Discovering radians in a whole circle (<i>SE p. 132</i>)
	10.3.2 Exploration (10 min.)	Discovering radian measure (<i>SE p. 131</i>)	10.3.4 Bring It Together (10-15 min.)	Connecting arc length to radius (<i>SE p. 133</i>)
	10.3.5 Wrap-Up (~5 min.)	Reflection on the learning of the lesson (<i>SE p. 134</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Radian <u>Additional Glossary:</u> - Arc Length		N/A	10.3.2 Exploration contains a Desmos activity to accompany the question prompts.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may harbor the misconception that a radian measure must be a multiple of π. - Students may have difficulty simplifying fractions to see the connection between the ratios of the circumference to the given radii. - Students may use the area formula during 10.3.3 Exploration instead of the circumference formula.	- Consider having large posters with differently sized circles drawn on them, and have students measure the radius and arc length using a ruler and a string. - Students will not be converting from degrees to radians yet.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect This lesson provides students the opportunity to discover radians. In 10.3.2 Exploration, encourage students to compare and connect the central angle, arc length, and radius. Allow students to discuss with other students to strengthen their understanding.	MA.K12.MTR.5.1: Patterns and Structure Students will be discovering the connection between the radius, central angle, and arc length. Use this as an opportunity for students to explore the patterns and structures to be able to conjecture the relationships.
 Differentiate Instruction	10.3.2 Exploration provides both visual and kinesthetic entry points for learners to engage with the content. The tables in this exploration provide additional support for visual learners to see the connection between the radius, central angle, and arc length. Engaging the class in discussions by having students share observations they made and explain their discoveries will help auditory learners deepen their understanding.	
Lesson Notes:		

10.3.1 Warm-Up: Would You Rather?

Component Overview: Students will identify which option they would rather have. Students will be asked to explain their choice using mathematics.



10.3.1 Warm-Up: Would You Rather?

Would you Rather...		
Deposit \$5 in the bank and have it triple each week for 4 weeks?	OR	Deposit \$3 in the bank and have it quadruple each week for 4 weeks?

- Which would you rather choose? Justify your reasoning using mathematics.

Answers will vary. Putting \$5 in the bank to triple for 4 weeks would give me \$405, whereas putting \$3 in the bank and quadrupling it for 4 weeks would give me \$768. So, I would rather deposit \$3 in the bank and have it quadruple each week for 4 weeks.



Do you think the two options will ever equal the same amount of money? Why or why not?

10.3.2 Exploration: Discovering Radian Measure

Component Overview: In this Exploration, students will work with Desmos to explore the impact the radius measure has on the arc length and the central angle. Students will be able to conclude that radians are based on the number of radii wrapped around the circumference of a circle, and the radian measure of a central angle is equal to the length of the arc intercepted by the central angle divided by the radius of the circle.



10.3.2 Exploration: Discovering Radian Measure

Angles that are measured in degrees can also be measured in **radians**. Measuring in radians is similar to measuring distance using other units, such as inches or feet. Radians can be written as a decimal or in pi (π) form.

With your partner, work through the Desmos activity "Discover Radian Measure." Complete the steps to answer each question.

Take a moment to look over the activity and the information provided.

- Slider " r " adjusts the radius of the circle. The radii for the activity range from 0 to 3, increasing by 0.5.
- Slider " c " adjusts the central angle.
- The central angle, $C_{entralAngle}$, is measured in radians, written as a decimal. Radians for the activity increase by 0.25.

Drag slider r to the right and left to observe how the measurements of radius, central angle, and arc length change as the r value changes. These values are in rows 2, 5, and 6 respectively of the activity.

Hit play on slider c to watch how the measurements of radius, central angle, and arc length changes. These values are in rows 2, 5, and 6 respectively.

1. Adjust slider c , $C_{entralAngle}$, so that the arc length, $A_{rcLength}$, equals the radius, r . Complete the table for three different measures of r .

Answers will vary.

Radius (r)			
$C_{entralAngle}(c)$			
$A_{rcLength}$			



The 10.3.2 Exploration can be done in pairs, in groups, or as a whole class. If technology is limited, consider projecting the activity for the class and play the sliders. Make the adjustments as prompted by the activity.



Consider splitting students into groups, with some groups exploring the impact of radius measure on arc length, and the other groups exploring the impact of angle measure on the arc length. Have students come back together and encourage students to explain their findings and think about the variation among arc length, central angle, and radius.

10.3.2 Exploration: Discovering Radian Measure (continued)

2. Describe the relationship observed in the table between the measure of the arc length and the measure of the radius.

The measure of the arc length is only equal to the radius if the central angle is measured to be 1 radian. Arc length equals the central angle multiplied by the radius.

3. Adjust slider r , radius, so that the arc length, $A_{rclength}$, is twice the radius, r . Complete the table for three different measures of r .

Answers may vary.

Radius (r)			
CentralAngle (c)			
$A_{rclength}$			

4. Describe the relationship observed in the table between the measure of the arc length and the measure of the radius.

Answers may vary. The arc length is only double the radius if the central angle measures 2 radians. When the central angle measures 2 radians, the arc length will be double the radius.



Consider asking the following:

- How does the radius vary with the arc length?
- How does the radius vary with the central angle?
- Is the relationship direct or inverse variation?

10.3.2 Exploration: Discovering Radian Measure (continued)

5. Complete the statements based on your work in questions 1-4.

A radian is a unit used to express the measure of an

angle. Radians are based on the number of ☐ angles
☐ radii

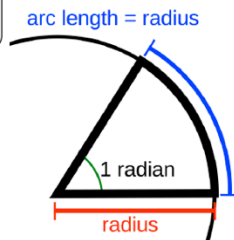
of a circle being wrapped along the ☐ diameter
☐ circumference

of the circle. The radian measure of a central angle is

equal to the length of the ☐ arc
☐ radius

intercepted by the central angle divided by the

☐ radius.
☐ arc length.



To help students understand the relationship between the radius and the arc length, consider having students do a hands-on activity:

- Draw a circle with a piece of string and a pencil.
- Place the string on the circumference of the circle.
- Use the end points on the string to create a central angle.

Ask students, “What do you notice about the arc length? How is it related to the radius?”

6. Describe how the diagram shows the relationship between the radius and arc length.

Answers will vary. It shows that the central angle (in radians) measure equals the $\frac{\text{arc length}}{\text{radius}}$. The ratio of the arc length to the radius will be equal to the central angle (measured in radians).

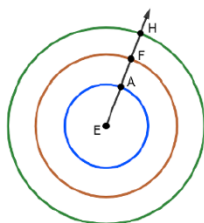
10.3.3 Exploration: Discovering Radians in a Whole Circle

Component Overview: Using different radii, students will discover that the ratio of the circumference to the radius will equal 2π , which is the radian measure of a whole circle.



10.3.3 Exploration: Discovering Radians in a Whole Circle

- Consider the diagram depicting three concentric circles each with center E . Point A lies on the blue Circle E , and has radius $EA = 1$ unit. Point F lies on the brown Circle E , and has radius $EF = 2$ units. Point H lies on the green Circle E , and has radius $EH = 3$ units.



- Complete the table with your partner.

Circle	Circumference	Ratio of Circumference to Radius
Radius $EA = 1$ unit	2π	$\frac{2\pi}{1} = 2\pi$
Radius $EF = 2$ units	4π	$\frac{4\pi}{2} = 2\pi$
Radius $EH = 3$ units	6π	$\frac{6\pi}{3} = 2\pi$

- Discuss any observations you made about the values in the table with your partner. Summarize your discussion.
The ratio of the circumference to the radius of each Circle E color is equal.

- Write a conjecture about the ratio of the circumference to radius and a whole circle measured in radians.
The ratio of the circumference to the radius of any whole circle will be equal to 2π , which is the radian measure of any whole circle.



Consider asking the following:

- Do you think the ratio of the circumference to the radius will always be 2π ?
- If 2π is the radian measure of a whole circle, what do you think would be the radian measure of half a circle? What about a quarter of a circle?

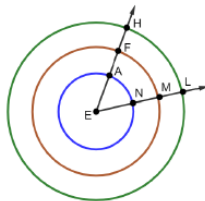
10.3.4 Bring It Together: Relating Arc Length to the Radius

Component Overview: Students will use concentric circles to conclude the proportional relationship among arc length and radius.



10.3.4 Bring It Together: Relating Arc Length to the Radius

- Concentric circles with center E are shown. Points A and N lay on the blue Circle E , and has radius $EA = 1$ unit. Points F and M lay on the brown Circle E , and has radius $EF = 2$ units. Points H and L lay on the green Circle E , and has radius $EH = 3$ units.



Describe how the arclengths AN , FM and HL are related.

Answer will vary.

The arc lengths are proportional.

For instance, $\frac{AN}{1} = \frac{FM}{2} = \frac{HL}{3}$.

Which is the same as: $AN = AN, FM = 2AN, HL = 3AN$

- Use words from the bank below to complete the statements.

angle	arc	radian	radius
-------	-----	--------	--------

The length of the **arc** intercepted by a(n) **angle** is proportional to the **radius** of the circle. The **radian** measure of the angle is the constant of proportionality.



$$\text{Radians} = \frac{\text{Circumference}}{\text{Radius}} = \frac{2\pi r}{r} = \frac{2\pi}{1}$$

$$360^\circ = 2\pi \text{ radians}$$

$$180^\circ = \pi \text{ radians}$$



Notice and Wonder

Consider providing students with the opportunity to write down things they notice and wonder about the concentric circles. This will allow students the chance to make observations before answering the question.

10.3.5 Wrap-Up: Cool Down

Component Overview: Students will summarize what they learned from the lesson by writing a note to help an absent student understand the relationship between the length of an arc intercepted by an angle and the radius.



10.3.5 Wrap-Up: Cool Down

1. Your friend was absent today and will need notes from class. Write a quick note to help them understand relationship between the length of an arc intercepted by an angle and the radius.

Answers will vary.

The arc length intercepted by an angle is proportional to the radius of the circle.

The radius multiplied by the central angle (measured in radians) will equal the arc length.

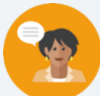


Consider having students reflect and document the challenges they faced in this lesson as well as what they learned. They can use their notes to prepare for future lessons.

10.4 Solving Problems Involving Arc Length and Area of a Sector

Lesson Introduction

 Benchmark	MA.912.GR.6.4 Solve mathematical and real-world problems involving the arc length and area of a sector in a given circle.		 Learning Target	I can solve mathematical and real-world problems involving arc length and sector area.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- How can you derive the formula used to find the arc length of a sector using the central angle and circumference of a circle? - How can you derive the formula used to find the area of a sector using the central angle and circumference of a circle?			
 Lesson Narrative	In this lesson, the student will be finding the area of a sector and the length of an arc, respectively, as a fraction of the area and the circumference of the entire circle, using the ratio between the central angle measure and the measure of a whole circle.			
 Component Summary	10.4.1 Warm-Up (~5 min.)	Comparing two deals of pizza prices (<i>SE p. 135</i>)	10.4.4 Collaboration (10 min.)	Discovering the arc length formula (<i>SE p. 137</i>)
	10.4.2 Exploration (10 min.)	Discovering the area of a sector formula (<i>SE p. 136</i>)	10.4.5 Guided Practice (5 min.)	Finding arc length (<i>SE p. 138</i>)
	10.4.3 Guided Practice (5-10 min.)	Solving using area of a sector (<i>SE p. 137</i>)	10.4.6 Guided Practice (5 min.)	Solving using area of a sector and arc length (<i>SE p. 139</i>)
	10.4.7 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p.140</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Area of a Sector - Arc Length <u>Additional Glossary:</u> N/A		(Optional) Large Posters	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may have difficulty isolating the central angle in 10.4.3 Guided Practice.	- Consider having a student write the formulas for finding the area and circumference of a circle on the board for students to use as a reference throughout the lesson. - Consider creating an anchor chart with students as they learn the formulas for Sector Area and Arc Length.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 10.4.2 Exploration positions students to notice the effects of a central angle on the area of the sector in the table provided. Consider applying this routine with different measures of central angles to discover the relationship between the central angle and the area of the sector.	MA.K12.MTR.6.1: Reasonableness Students are required to perform multiple steps to find the area of a sector and the arc length. Due to the problems requiring students to use the order of operations, students may make minor errors that result in an incorrect solution. Encourage students to always check each step of their work, and ask students to verify their solutions by explaining their work.
 Differentiate Instruction	Read and discuss problems aloud to assist students who benefit from auditory entry points. Encourage students to use visuals, such as drawing out circles for students who prefer visual learning methods. Consider allowing students to create anchor charts on larger posters to help kinesthetic and visual learners.	
Lesson Notes:		

10.4.1 Warm-Up: Bell Work

Component Overview: Students will choose between two differently sized pizzas and justify their preferred choice.



10.4.1 Warm-Up: Bell Work

1. A small, family-owned restaurant sells a personal pan pizza with a radius of 10 centimeters for \$5. A nearby restaurant cuts their pizza with a radius of 30 centimeters into 8 slices of equal size, and charges \$5 per slice.

Which is a better deal? Explain your reasoning using mathematics.

Answers may vary. The personal pan pizza has a total area of $100\pi \text{ cm}^2$ for \$5. Meanwhile, the nearby restaurant makes pizzas with an area of $900\pi \text{ cm}^2$. This means that the area of one slice is equal to $\frac{900\pi}{8} = 112.5\pi \text{ cm}^2$. So, for \$5, with an area of $112.5\pi \text{ cm}^2$, the slice is the better deal since you will get more pizza for the same price.



Consider prompting students with questions to help them determine which one is a better deal:

- What is the area of the personal pan pizza?
- What is the area of one slice of the 30 centimeter pizza?
- Would you rather have a pizza with a smaller or larger area?

10.4.2 Exploration: Finding a Formula for the Area of a Sector

Component Overview: Students will work with a partner to discover the formula for the area of a sector by making observations of the fraction of a circle, the central angle, and the equation that can be used to find the area of the sector given.



10.4.2 Exploration: Finding a Formula for the Area of a Sector

- Suppose a circle is cut into 8 equal slices to form sectors.

- Complete the table.

Number of Slices	Fraction of the Circle	Central Angle in Degrees ($^{\circ}$)	Write an Equation to Find the Area of the Sector
2	$\frac{2}{8} = \frac{1}{4}$	$\frac{360^{\circ}}{4} = 90^{\circ}$	$A = \frac{\pi r^2}{4}$
4	$\frac{4}{8} = \frac{1}{2}$	$\frac{360^{\circ}}{2} = 180^{\circ}$	$A = \frac{\pi r^2}{2}$
6	$\frac{6}{8} = \frac{3}{4}$	$\frac{3 \cdot 360^{\circ}}{4} = 270^{\circ}$	$A = \frac{3\pi r^2}{4}$
8	$\frac{8}{8} = 1$	$1 \cdot 360^{\circ} = 360^{\circ}$	$A = \pi r^2$

- Compare the values in your table with your partner. Discuss your equations in the last column and make any revisions if needed.

Students do not need to write an explanation here, but a sample response is provided for reference. For 6 slices, I did not multiply by 3, I only divided by 4.

- Discuss with your partner how the fractions in your equations relate to the central angles, and how the equations might have differed if the angle measure was given in radians.

Students do not need to write an explanation here, but a sample response is provided for reference. The fractions in my equations for the area of the sector are the same as the fraction of the circle's area represented by the central angle, in degrees. If the angle was given in radians, the fraction for the area of the sector would be $\frac{1}{2}$ times the radian measure.



Notice and Wonder

Consider providing students with the opportunity to write down things they notice and wonder about the table. This will allow students the opportunity to make observations before discussing with a partner.



Encourage students to think of both ways of writing a general formula to find the area of any sector by using degrees and radians.

- What is another unit of measure for angles?
- How could you rewrite your formula to include this unit of measure for angles?
- How would you know when to use which formula?

10.4.2 Exploration: Finding a Formula for the Area of a Sector (continued)

- d. Summarize your discussion by writing a general formula to find the area of any sector.

Answers may vary.

Area of Sector = $\frac{\text{central angle}}{360^\circ} \cdot \pi r^2$, when in degrees.

OR Area of Sector = $\frac{\text{central angle}}{2} \cdot r^2$, when in radians.



Area of a Sector

$$A = \frac{\theta}{360} \pi r^2, \text{ where } \theta \text{ is in degrees}$$

$$A = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$

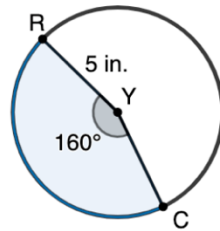
10.4.3 Guided Practice: Area of a Sector

Component Overview: Students will practice finding the area of a sector using the formula.



10.4.3 Guided Practice: Area of a Sector

1. Circle Y has radius $\overline{CY}$ that measures 5 inches with central angle $\angle RYC$, where $m\angle RYC = 160^\circ$. Find the area of the shaded sector. Round your answer to the nearest thousandth.



$$A = \frac{\theta}{360^\circ} \cdot \pi r^2$$

$$A = \frac{160}{360} \cdot \pi (5)^2 = 11.111\pi \text{ in}^2$$

$$\approx 34.907 \text{ in}^2$$

2. Circle X has a diameter of 12 centimeters. Points B and P are on the circle and form the sector BXP . The central angle $\angle BXP$ in Circle X measures 60° . Determine the area of the sector. Express your answer in terms of π .

$$A = \frac{\theta}{360^\circ} \cdot \pi r^2$$

$$A = \frac{60}{360} \cdot \pi (6)^2 = 6\pi \text{ cm}^2$$

3. The area of a sector with a radius of 8 inches is 74.84 square inches. Calculate the approximate measure of the central angle of the sector in degrees.

$$\text{Area of Sector} = \frac{\theta}{360^\circ} \cdot \pi r^2, \text{ with } \theta = \text{central angle.}$$

$$74.84 = \frac{\theta}{360} \cdot \pi (8)^2$$

$$\frac{74.84(360)}{64\pi} = \theta$$

$$\theta \approx 134.0^\circ$$



Consider asking the following:

- How could you find the radius if you were given the area of a sector and the measure of the central angle?
- For question 1, how could you find the area of the non-shaded sector?

10.4.4 Collaboration: Arc Length

Component Overview: Students will work with a partner to make connections with the central angle and the arc length to create a general equation that can be used to find the arc length of a sector.



10.4.4 Collaboration: Arc Length

Work with your partner to complete the following.

1. Explain how the circumference relates to an arc length of a circle.
Arc length is the portion of the circumference. The ratio of arc length to circumference is equal to the ratio of the central angle to the whole circle.
2. Describe how to find the arc length of half a circle.
First, find the circumference of the circle, then divide it by two.
3. Describe how to find the arc length of a quarter of a circle.
First, find the circumference of the circle, then divide it by four.
4. Describe how to find the arc length of a circle with a central angle of 60° .
Find the ratio of the central angle to the whole circle and then multiply that ratio by the circumference of the circle.
5. Write a general equation that can be used to find the arc length of a sector in degrees.
Arc Length = $\frac{\text{central angle}}{360^\circ} \cdot 2\pi r$



For visual learners, consider providing students with a circle with a given circumference. Have students relate the questions to the circle and circumference they were given to help students make connections between the central angle and arc length.

10.4.4 Collaboration: Arc Length (continued)

6. Discuss with your partner how your equation would change if the central angle had been given in radians. Summarize your discussion by writing an equation to find the arc length if the angle measure is given in radians.

I would multiply the radius by the central angle.

I would convert 360° to its radian measure of 2π , then I would divide and cancel out the two 2π 's, leaving the equation:

Arc Length = central angle $\cdot$ r , with the central angle now measured in radians.



Arc Length

Arc Length = $C \cdot \frac{\theta}{360^\circ}$, where θ is in degrees

Arc Length = $r \cdot \theta$, where θ is in radians



For visual learners, consider providing students with a circle with a given circumference. Have students relate the questions to the circle and circumference they were given to help students make connections between the central angle and arc length.

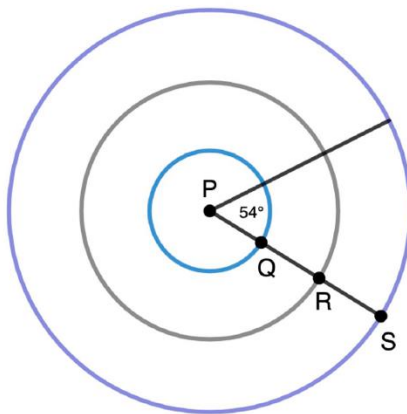
10.4.5 Guided Practice: Arc Length

Component Overview: Students will practice finding the arc length of a sector using the formula.



10.4.5 Guided Practice: Arc Length

- Concentric circles with center P are shown with $m\angle P = 54^\circ$. The smallest Circle P has a radius $\overline{PQ}$ that measures 15 meters (m). The largest Circle P has radius $\overline{PS}$ and the middle Circle P has radius $\overline{PR}$. Each circle has a radius that is 15 m longer than the immediate prior circle.



Calculate the arc length intercepted by $\angle P$ on each circle.

Arc length of circle with radius $\overline{PQ}$	$\begin{aligned} \text{Arc Length} &= \frac{54}{360} \cdot 2\pi(15) \\ &= 4.5\pi \\ &\approx 14.14m \end{aligned}$
Arc length of circle with radius $\overline{PR}$	$\begin{aligned} \text{Arc Length} &= \frac{54}{360} \cdot 2\pi(30) \\ &= 9\pi \\ &\approx 28.27m \end{aligned}$
Arc length of circle with radius $\overline{PS}$	$\begin{aligned} \text{Arc Length} &= \frac{54}{360} \cdot 2\pi(45) \\ &= 13.5\pi \\ &\approx 42.41m \end{aligned}$



Consider asking the following:

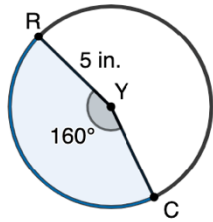
- How are the arc lengths of each sector different if they have the same central angle?
- How could you find the central angle if you were given the radius and the arc length?



Consider challenging students to calculate the arc length in terms of π also.

10.4.5 Guided Practice: Arc Length (continued)

2. Points R and C are on Circle Y . If $m\angle CYR$ is 160° , then find the length of $\widehat{RC}$.



$$\begin{aligned}
 \text{Arc Length} &= \frac{160}{360} \cdot 2\pi(5) \\
 &= \frac{4}{9} \cdot 10\pi \\
 &= \frac{40}{9}\pi \\
 &\approx 13.96 \text{ in}
 \end{aligned}$$



If students found the length of arc RC to be approximately 34.9 in., then they used the area of the circle instead of the circumference.

10.4.6 Guided Practice: Solving Real-World Problems Involving Arc Length and Area of a Sector

Component Overview: Students will calculate arc length and the area of a sector in real-world context.



10.4.6 Guided Practice: Solving Real-World Problems Involving Arc Length and Area of a Sector

1. In Sarah's family, the person celebrating their birthday always gets to cut the first piece of a circular birthday cake. Sarah is celebrating her birthday with both of her parents, her two brothers and her best friend. She cuts her piece of birthday cake and it formed a sector with a radius of 8 inches and a central angle with measure 85° . If the 5 other party guests equally share the remaining portion of the cake, what is the approximate area of cake each person receives? Round your answer to two decimal places.

First, find the area of the entire cake:

$$\text{Area of Cake} = \pi \cdot 8^2 = 64\pi.$$

Next, find the area of the section of cake cut out by Sarah:

$$\text{Area of Sector} = \frac{85 \cdot 8^2 \cdot \pi}{360} = \frac{136}{9}\pi \text{ in}^2$$

Subtract to get the remaining area:

$$\text{Remaining Area} = 64\pi - \frac{136}{9}\pi = \frac{440}{9}\pi \text{ in}^2$$

Finally, divide the remaining area by 5:

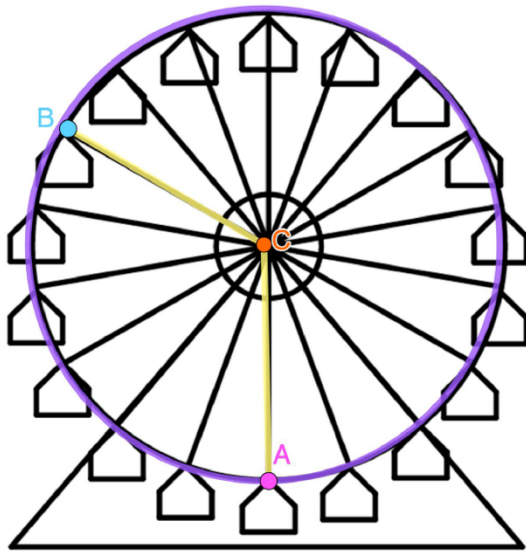
$$\begin{aligned} \text{Area of each piece of cake} &= \frac{440}{9}\pi \div 5 \\ &= \frac{88}{9}\pi \text{ in}^2 \text{ Area of each piece of cake} \approx 30.72 \text{ in}^2 \end{aligned}$$



For students who struggled during the lesson, consider using this time to work with individual or small groups of students. It will be important to determine which part in the process students may be struggling with. Consider having students start the problem to identify where the help is needed.

10.4.6 Guided Practice: Solving Real-World Problems Involving Arc Length and Area of a Sector (continued)

2. Dominic is riding a Ferris wheel at the county fair. The Ferris wheel forms a circle with center C . The Ferris wheel has a radius of 60 ft and rotates clockwise with seats that are equidistant from each other.



Consider reminding students:

- Which way clockwise is.
- That a revolution is one full circle.

- a. After the ride begins from point A it pauses at point B . How many feet has Dominic traveled when his seat pauses at point B ? Write your answer in π form and rounded to the nearest thousandth.

π or feet

- b. About how many feet has Dominic traveled if the full ride is 4 revolutions?

π or feet

10.4.7 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



10.4.7 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Answers will vary.

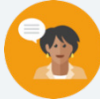


Consider providing online resources for additional enrichment opportunities for students to practice or test their abilities.

10.5 Solving Problems Involving Arc Measure and Central Angles

Lesson Introduction

 Benchmark	MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.		 Learning Target	I can use the relationship between the measure of a central angle and its intercepted arc to solve problems.
 Benchmark Coherence	Building On			Working Toward
	N/A			N/A
 Essential Questions	How can I use the relationship between the central angle measure and its intercepted arc to solve problems?			
 Lesson Narrative	In this lesson, students will explore central angles to discover the relationship between the arc measure and central angles. After exploration, instruction, and practice with the concept, students will then apply those concepts to solve mathematical problems involving central angles and intercepted arcs.			
 Component Summary	10.5.1 Warm-Up (~5 min.)	Noticing polygons inscribed in circles (<i>SE p. 141</i>)	10.5.4 Guided Practice (10-15 min.)	Solving problems involving central angles and intercepted arcs (<i>SE p. 145</i>)
	10.5.2 Exploration (10 min.)	Discovering the relationship between central angles and intercepted arcs (<i>SE p. 142</i>)	10.5.5 Your Turn! (5-10 min.)	Solving problems involving central angles and intercepted arcs (<i>SE p. 146</i>)
	10.5.3 Bring It Together (5 min.)	Formalizing the relationship between central angles and intercepted arcs (<i>SE p. 144</i>)	10.5.6 Wrap-Up (~5 min.)	Finding the measure of the central angle (<i>SE p. 147</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Arc - Arc Length - Arc Measure - Major Arc - Minor Arc <u>Additional Terms:</u> N/A		(Optional) Straight Edge (Optional) Markers/Colored Pencils (Optional) Compass	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may think the measure of an arc is half the measure of the central angle that intercepts the arc. - Students may work 360° into their work based on what they recall about circles. - Students may not recognize different arcs when overlap is used in a problem.	10.5.2 Exploration is designed to be a discovery activity for students. Be cautious not to give away the formalized information prior to the 10.5.3 Bring It Together.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 10.5.2 Exploration provides an opportunity to leverage the Think-Pair-Share routine. In question 1, part j, students will work with a partner to share their explanations of the work they have done independently on question 1 parts a – i. This move will build confidence prior to any whole-class discussion on the relationship of central angles and intercepted arcs.	MA.K12.MTR.4.1: Discussion and Reflection In 10.5.2 Exploration, students get to work together to understand the relationship between central angles and intercepted arcs. During this activity they will work to communicate mathematical ideas, which is a component of this MTR Standard.
 Differentiate Instruction	Consider creating an anchor chart for 10.5.3 Bring It Together when students bring together their exploration and the formal introduction to the relationship between central angles and intercepted arcs. There are several different angle relationships that students learn about in this unit. Consider leveraging anchor charts to assist students with recalling the relationships based on the different components.	
Lesson Notes:		

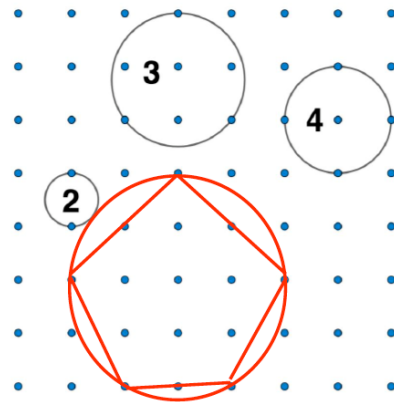
10.5.1 Warm-Up: Circles

Component Overview: Students will work with three given circles to share what they notice, and then create a polygon inscribed in a circle.



10.5.1 Warm-Up: Circles

1. A square lattice with three circles is shown where the circles pass through exactly 2, 3, or 4 points on the lattice, respectively.



- a. What do you notice about the circles?

Answers will vary. Each circle is a different size. A larger number does not correspond to a larger circle. The points evenly divide each circle into congruent arcs.

- b. Connect the points on each circle to create a polygon. Describe any pattern you notice.

Answers will vary. The line segments within each circle appear to have the same length.

- c. What type of polygon would be inside a circle that passes through 5 points?

Pentagon (5 sides)

- d. On the lattice, draw a circle that passes through exactly 5 points.

Answers will vary. On lattice.



Consider leveraging a Think-Pair-Share routine with this activity. Allow students to have some individual think time in order to try it on their own. Then, allow them to work with a partner to share their thinking and strategy. Allow student pairings to share out their answers to the entire group at the conclusion of this activity.

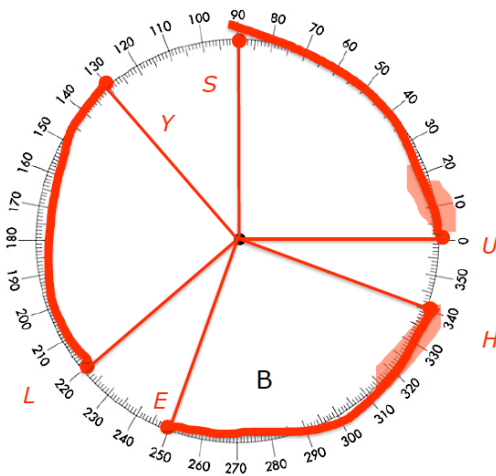
10.5.2 Exploration: Understanding Central Angles

Component Overview: Students will explore central angles using two, multi-part, scaffolded question series. For this activity, students will start working independently before sharing their answers with a partner.



10.5.2 Exploration: Understanding Central Angles

1. Circle B is shown.



- a. Find the 130° mark and label it point Y . Then, find the 220° mark and label it point L . Draw along the circumference from point Y to point L going counterclockwise.

Shown on diagram.

- b. Exactly what fraction of the circle's circumference is from point Y (130°) to point L (220°)?

$$\frac{1}{4}$$

- c. Draw the radius $\overline{BY}$ from the center B to point Y . Then, draw another radius $\overline{BL}$ from the center B to point L .

Answer on diagram.

- d. What is the measure, in degrees, of the central angle $\angle YBL$ in relation to $\widehat{YL}$?

$$90^\circ$$



Students do not need tools in order to complete this exploration. Having a straightedge and compass may be beneficial for students when they have to draw lines and arcs, although they are not constructing them. It can be helpful to keep the lines straight to avoid any visual confusion.



How did you determine the fraction of the circle's circumference in question 1 part b?

10.5.2 Exploration: Understanding Central Angles (continued)

- e. Next, along the circumference, label 0° as point U , 90° as point S , 250° as point E , and 340° as point H .

Answer on diagram.

- f. Draw two separate arcs along the circumference of the circle, one extending from point U to point S and the other extending from point E to point H . Then, draw the central angles that intersect with each part of the circumference you marked.

Answer on diagram.

- g. What is the measure, in degrees, of each of the central angles, $\angle UBS$ and $\angle EBH$?

90°

- h. Describe what you notice about all three central angles.

All three central angles are congruent.

- i. Explain the relationship between a central angle and its intercepted arc.

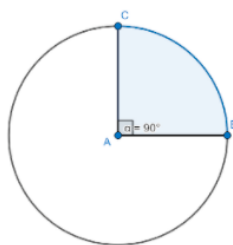
A central angle and the arc it intercepts have the same degree measure.

- j. Share your response with your partner. Listen to your partner's explanation and write a summary of the explanation. Have your partner initial stating that your summary of their explanation is accurate.

Your Partner's Explanation	Partner's Initials
<i>Answers will vary.</i>	

- k. Draw a diagram that shows the relationship between a central angle and its intercepted arc.

$m\angle CAB = 90^\circ$ and $m\widehat{CB} = 90^\circ$



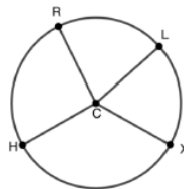
When circulating and monitoring student work, listen intently to student discussions on question 1 part j. If students are working on question 1 part k or beyond, prompt them to respond to question 1 part j in order to hear their articulation, as it is very important prior to the formal angle relationship introductions in the next activity.

10.5.2 Exploration: Understanding Central Angles (continued)

1. Discuss with your partner how arc measure relates to the relationship between a central angle and its intercepted arc. Summarize your discussion.

Answers will vary. Since a central angle represents a fraction of a circle, we know that the intercepted arc measure is the same fraction of the total circumference.

2. Circle C is shown where points R , L , X , and H are on the circle.



- a. Name the angle that intercepts $\widehat{RH}$.
 $\angle RCH$
- b. Name the arc that is intercepted by $\angle HCX$.
 $\widehat{HX}$

- c. Brooklynn and Angelina were discussing how to name an arc that is intercepted by $\angle RCX$. Each of their explanations are shown.

Brooklynn	Angelina
The arc that is intercepted by $\angle RCX$ is $\widehat{RX}$ because you just use the first and last letters of the angle to name the arc.	An arc that is intercepted by $\angle RCX$ could be $\widehat{RLX}$ or $\widehat{RHX}$ because the angle intercepts two arcs.

Discuss with your partner the conversation between Brooklynn and Angelina. Summarize your discussion.

Answers will vary. Brooklyn and Angelina agree that an arc's name must include the two points on the circle of the angle that intercepts it. Angelina notices that the measure of the angle could be either less than or greater than 180° , so there are two possible arcs that may be intercepted. Including three points in the arc's name denotes which arc is being referred to. Neither girl defined which arc is being referred to if only two points are named.



When monitoring student conversation and summaries, listen for and encourage the use of proper mathematical vocabulary.



When concluding 10.5.2 Exploration, engage students in a whole-group discussion about their response to question 2 part c. This will help highlight any misconceptions when naming arcs, which is important before moving into future lessons.

10.5.3 Bring It Together: Arc Measure and Central Angles

Component Overview: Students will be formally introduced to the relationship of a measure of an arc, and the measure of the central angle that intercepts that arc.



10.5.3 Bring It Together: Arc Measure and Central Angles

An **arc** of a circle is a section of the circumference of the circle.



The **length of an arc** is the length of the section of the circumference.



In a circle, the measure of an arc (in degrees or radians) is equal to the measure of the central angle that intercepts the arc.



In a circle, any arc whose corresponding central angle has a measure greater than 180° is considered a **major arc** and should be named using three letters.



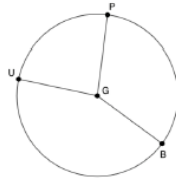
In a circle, any arc whose corresponding central angle has a measure less than 180° is considered a **minor arc** and should be named using two letters.



To provide a visual component, consider having students complete a graphic organizer to include pictorial representations of the information included in the blue boxes of this section.

10.5.3 Bring It Together: Arc Measure and Central Angles (continued)

1. Circle G has points P , B , and U on the circle with $m\angle PGB = 120^\circ$ and $m\widehat{BU} = 155^\circ$. Ryan and Katelyn are finding the measure of $\widehat{UP}$, but they found different answers.



Ryan	Katelyn
$155 + 120 = 275$	$155 + 120 = 275$
$275 - 180 = 95$	$360 - 275 = 85$
$m\widehat{UP} = 95^\circ$	$m\widehat{UP} = 85^\circ$

Do you agree with Ryan or Katelyn? Justify your answer.

Answers will vary. I agree with Katelyn because the measures of all three arcs should add to 360° in order to complete the entire circle, so Katelyn correctly subtracted the total measures of the two given arcs in order to find the third.



Question 1 is designed to demonstrate what students discovered in 10.5.2 Exploration. Use this opportunity to address student misconceptions, and ensure they have a solid understanding of the angle relationships provided.

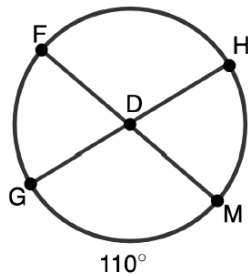
10.5.4 Guided Practice: Finding Angle Measures

Component Overview: Students will practice finding angle measures using what they have learned about central angles and intercepted angle relationships.



10.5.4 Guided Practice: Finding Angle Measures

1. Circle D has diameters $\overline{FM}$ and $\overline{GH}$ that both intersect at Point D . Given the $m\widehat{MG} = 110^\circ$, find the $m\widehat{HF}$.



110°

2. Circle T has points A , V , and C on the circle with $m\angle VTA = 21.7^\circ$. If $m\widehat{VC} = 132.3^\circ$ and $\widehat{VC}$ does not intersect $\widehat{VA}$, what is $m\angle CTA$?
3. Circle C has radii $\overline{CN}$ and $\overline{CL}$. Find the value of x if $m\angle LCN = 114^\circ$ and $m\widehat{LN} = (6x + 6)^\circ$.
4. Isabella is creating a folding fan for art class. She needs to determine the measure of the arc of the fan before she can cut the paper.

She uses the expression $(8x + 102)^\circ$ as the measure of the central angle, $\angle NFG$. For the $m\widehat{GN}$, she uses the expression $(32x + 54)^\circ$.



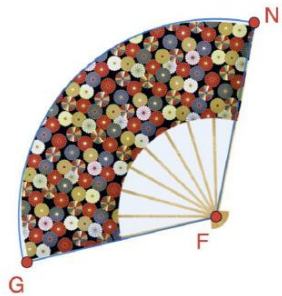
Students will be applying the relationships they learned in the prior activities to solve problems. Consider modelling question 1 for students, and sharing strategic thinking on how to approach the problem, including what information and relationships can be determined based on the information provided.



For questions 2 and 3, encourage students to draw out the circle as a visual aid when the visual is not provided in the question.

10.5.4 Guided Practice: Finding Angle Measures (continued)

What is the degree measure of the arc of the fan, $\widehat{GN}$?



$$(8x + 102)^\circ = (32x + 54)^\circ$$

$$2 = x$$

$$\widehat{GN} = (32(2) + 54)^\circ$$

$$= 118^\circ$$



Consider finding an actual fan to enhance this activity. If available, allow students to explore this problem using a similar fan to the one provided in question 4. This can create a stronger level of engagement with this activity.

To see a visual connection to the whole circle, have students use a compass to complete the drawing of the circle created by the radius of the fan.

10.5.5 Your Turn!

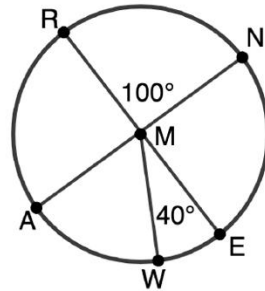
Component Overview: Students will complete two independent practice problems on solving problems using central angle relationships.



10.5.5 Your Turn!

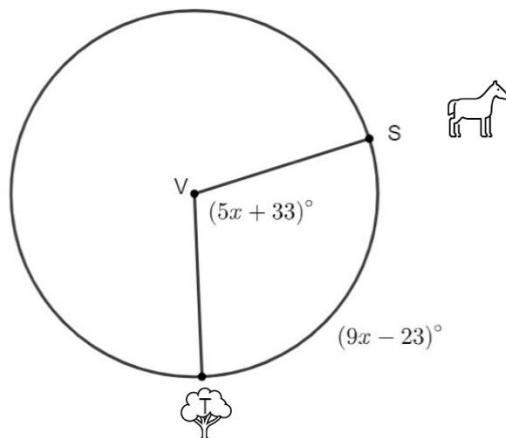
1. In Circle M , $m\angle RMN = 100^\circ$ and $m\angle EMW = 40^\circ$. Calculate the $m\widehat{RNW}$ given that M is the center and $\overline{RE}$ and $\overline{AN}$ are diameters.

$$m\widehat{RNW} = 220^\circ$$



2. Spirit is tied to a post at point V and is grazing in a circular pattern. He decides to walk from his position at point S to the tree at point T along the circumference of the circular area he can roam.

If the $m\angle SVT = (5x + 33)^\circ$ and $m\widehat{ST} = (9x - 23)^\circ$, find the measure of the arc that Spirit walked, $m\widehat{ST}$.



$$m\widehat{ST} = 103^\circ$$



For students who struggled during the previous component, consider using this time to work with individual or small groups of students on angle relationships. It will be important to determine which part in the process students may be struggling with. Some students might not remember the angle relationships, and some might apply them incorrectly or make calculation errors.



For students who finish early, have them use the circle from question 1 to find the measure of the following arcs:

- $m\widehat{AW}$
- $m\widehat{RW}$
- $m\widehat{RNW}$
- $m\widehat{ERA}$
- $m\widehat{WNA}$

10.5.6 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door on using the relationship between central angles and their intercepted arcs to solve problems.



10.5.6 Wrap-Up: Ticket Out the Door

- Circle X has points R , P , and A on the circle. The measure of the central angle $\angle AXP = 50^\circ$, $m\widehat{RA} = 141^\circ$, and $\widehat{RA}$ does not intersect $\widehat{AP}$. Find $m\angle RXP$.

$$m\angle RXP = m\widehat{RP} = 360^\circ - (m\widehat{AP} + m\widehat{RA})$$

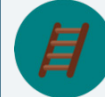
$$m\angle RXP = 360^\circ - (50^\circ + 141^\circ) = 169^\circ$$



Student performance on this Wrap-Up can provide valuable feedback on how well students understood the content of this lesson.

10.6 Solving Problems Involving Inscribed Angles

Lesson Introduction

 Benchmark	MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.		 Learning Target	I can use the relationship between the measure of an inscribed angle and its intercepted arc to solve problems.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Question	What is the relationship between an inscribed angle and the measure of the intercepted arc?			
 Lesson Narrative	In this lesson, students are introduced to the concept of inscribed angles. Students will begin by exploring the concepts of inscribed angles to discover the inscribed angles theorem. They will then apply that knowledge to solving mathematical problems that involve inscribed angles using algebraic expressions.			
 Component Summary	10.6.1 Warm-Up (~5 min.)	Determine measures of angles on a clock (<i>SE p. 148</i>)	10.6.3 Guided Practice (15 min.)	Solve problems using inscribed angles (<i>SE p. 150</i>)
	10.6.2 Exploration (10-15 min.)	Discover the relationship between an inscribed angle and its intercepted arc (<i>SE p. 149</i>)	10.6.4 Your Turn! (10 min.)	Solve problems using inscribed angles (<i>SE p. 152</i>)
	10.6.5 Wrap-Up (~5 min.)	Reflect on what is still confusing from the lesson (<i>SE p. 153</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Chord - Inscribed angle - Inscribed angle theorem <u>Additional Terms:</u> - Central angle - Arc measure - Intercepted arc		(Optional) Chart paper (Optional) Markers (Optional) GeoGebra (Optional) Protractor	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may think the inscribed angle is equal to the intercepted arc.	Do not give away anything too early in 10.6.2 Exploration, as it is designed to be a discovery activity where students discover the inscribed angle theorem.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share During 10.6.2 Exploration, students will have independent time to think through and complete one section of the activity (a-c), and then collaborate with their partner in part d to compare conclusions before moving to the final parts of the activity.	MA.K12.MTR.5.1: Patterns and Structure For 10.6.3 Guided Practice and 10.6.4 Your Turn!, students will be applying what they learned in the prior activity about inscribed angles to solve mathematical problems that involve algebraic expressions. To do this, students will need to focus on the relevant details within the problem, and decompose it into manageable parts.
 Differentiate Instruction	To increase student engagement, consider making parts of this lesson digital. For 10.6.2 Exploration, consider supporting the activity using the “Central Angles,” “Inscribed Angles,” and the “Intercepted Arcs” activity from GeoGebra or other digital resources.	
Lesson Notes:		

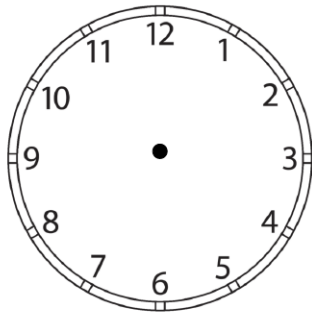
10.6.1 Warm-Up: Angles on a Clock

Component Overview: Students will explore the angles that are created by three different times of a standard clock.



10.6.1 Warm-Up: Angles on a Clock

1. Use the clock to answer the following questions.



- a. If the time is 10:10 and the hour hand does not move, what is the exact angle between the minute hand and hour hand?
120° or 240°
- b. If the hour hand does move, what is the exact angle if the time is 7:30?
45° or 315°
- c. If the hour hand does move, what is the exact angle if the time is 7:35?
17.5° or 342.5°



For this activity, consider providing students with a protractor to help them verify that their angle measures are correct after trying without one.



Some students will struggle with reading a standard clock. Be prepared to provide images of what 10:10 and 7:35 represent.

10.6.2 Exploration: Inscribed Angles

Component Overview: Students will explore the concept of inscribed angles through a series of two, multi-part, scaffolded questions. For this activity, students will start independently before sharing answers with a partner.



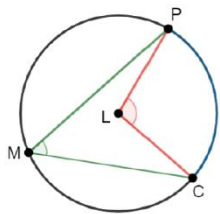
10.6.2 Exploration: Inscribed Angles

- Circle L is shown where P , C , and M are points on the circle.

A **chord** is a line segment on the interior of a circle with both endpoints lying on the circle.



An **inscribed angle in a circle** is an angle which is formed in the interior of a circle when two chords share an endpoint.



Central Angle $m\angle PLC = 110^\circ$

Inscribed Angle $m\angle PMC = 55^\circ$

Arc Measure $\widehat{PC} = 110^\circ$

- What is the connection between the central angle and the arc measure?

The measure of a central angle is equal to the measure of its intercepted arc.

- Describe the relationship between the central angle and the inscribed angle.

The measure of an inscribed angle is half that of its intercepted central arc.

- Write a conclusion about the measures of the central angle, the inscribed angle, and the arc measure.

Answers will vary. For an inscribed angle, a central angle, and an arc that all intercept the same two points on a circle the central angle and the arc have equal measures while the measure of the inscribed angle is half that of the central angle.



Consider creating a “parts of a circle” anchor chart where students can see all the different components with circles in one place. The chart should include radius, diameter, chord, arc, central angle, and inscribed angle. Only include vocabulary on this chart so that students can visualize the components, not all the relationships associated with each.

10.6.2 Exploration: Inscribed Angles (continued)

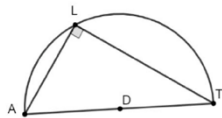
- d. Share your response to Part C with two classmates. Listen to their conclusion and summarize it in the table. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating your summary was correct.

My Summary of Their Conclusion	Initials
<i>Answers will vary.</i>	

- e. Using Circle L , complete the table.

Central Angle	Inscribed Angle	Arc Measure
216°	108°	216°
162°	81°	162°
143°	71.5°	143°
290°	145°	290°

2. $\angle ALT$ is inscribed in semicircle D , as shown.

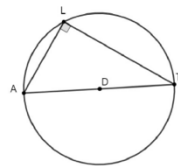


- a. Describe the angle inscribed in the semicircle.

$\angle ALT$ is a right angle

- b. If the entire Circle D is shown, what is the measure of $\widehat{AT}$?

$m\widehat{AT} = 180^\circ$



- c. Discuss with your partner the relationship between an inscribed angle and a semicircle.

An inscribed angle that intercepts the endpoints of a semicircle is a right angle.



Inscribed Angle Theorem

The measure of an inscribed angle is one-half the measure of the intercepted arc.



When circulating during this activity, listen for student responses to question 1 part d. This is a place where misconceptions, such as the relationship of the angle being the same instead of half, can be heard and addressed.



Consider asking the following for students stuck on question 1 part e:

- What do you know about the relationship between the central angle and the arc measure from the prior lesson? Consider filling in those answers first.
- What did you notice about the relationship between the inscribed angle and the arc measure?



Consider asking the following with question 2 part b:

- Is the arc created by the endpoints of a diameter a major arc or a minor arc? Explain your answer.
- How do we know which arc is being referred to when naming an arc created by a semi-circle?



Students may get stumped using the “semicircle” in question 2 part c. When circulating, remind students who are struggling to describe the relationship between a semicircle and a circle, and how that relationship impacts that inscribed angle.

10.6.3 Guided Practice: Inscribed Angles

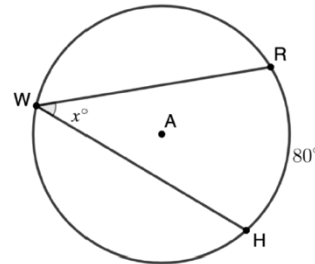
Component Overview: Students will complete four practice problems finding the missing measures and/or values using what they have learned about inscribed angles.



10.6.3 Guided Practice: Inscribed Angles

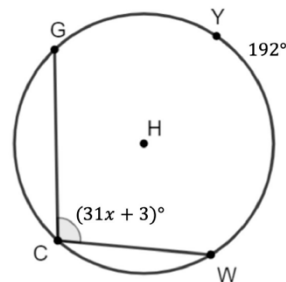
1. Circle A is shown where points R , W , and H are on the circle. $\angle RWH$ is inscribed inside Circle A with $m\widehat{RH} = 80^\circ$. Find the value of x .

$$x = 40$$

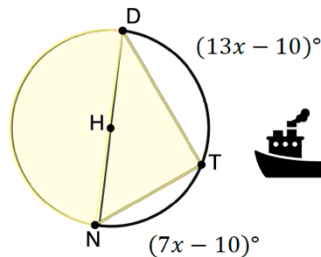


2. Points Y , G , C , and W lie on Circle H . $\angle GCW$ is inscribed in the circle. Find x , given $m\widehat{GYW} = 192^\circ$ and $m\angle GCW = (31x + 3)^\circ$.

$$x = 3$$



3. Boats use angle measures and arcs when sailing in treacherous waters. When a boat captain turns on the boat light, it forms inscribed angle $\angle NTD$. Use the information for Circle H to find $m\angle NDT$.



$$m\angle NDT = 30^\circ$$



For this activity, consider beginning with a teacher-modeled activity for question 1. In this activity, students will be applying what they have learned during the Exploration. Students can benefit from hearing structured thinking out loud on determining what is known from the problem, what relationships can be determined, and how those relationships can be used to solve problems.



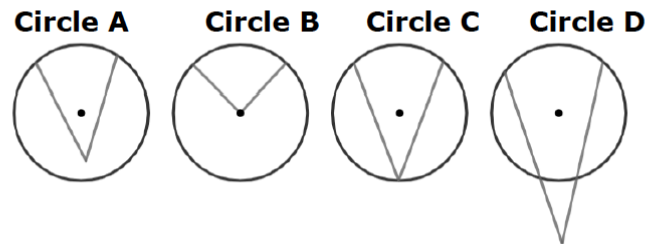
Students may struggle creating a diagram of the circle in question 2. Consider providing a circle with the points and angle drawn but with no labels as an entry point for those students into the problem.



Question 3 is the first real-world example with this content for this lesson. Some students may need assistance with pulling information from the problem, because they could be intimidated by the word problem.

10.6.3 Guided Practice: Inscribed Angles (continued)

4. Craig and Valentina are reviewing circles to identify inscribed angles.



Valentina identified Circle C is an inscribed angle. Craig identified both Circle A and Circle C as inscribed angles.

Do you agree with Valentina, Craig, or neither? Justify your reasoning.

Answers will vary. I agree with Valentina because inscribed angles are formed by two chords that intersect at the same point on the circumference of the circle.



Question 4 exhibits key features of MA.K12.MTR.4.1: Discussion and Reflection by having students analyze the mathematical thinking of others.



If students finish early, challenge them with writing a conjecture about the relationship of the angle in the image of Circle D to the arcs of the circle. Have them include a summary of their thinking.

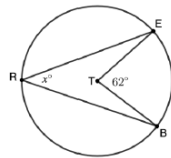
10.6.4 Your Turn!

Component Overview: Students will complete two independent practice problems with solving for angle measures and/or values using the inscribed angle relationships.

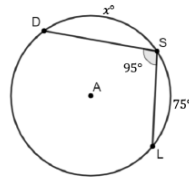


10.6.4 Your Turn!

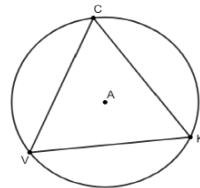
- Circle T is shown where $\angle ETB$ is a central angle, $\angle ERB$ is inscribed in the circle, and $m\angle ETB = 62^\circ$. Find $m\angle ERB$.
 $x = 31^\circ$



- Circle A has points D , S , and L on the circle. $\angle DSL$ is inscribed in Circle M measuring 95° . Given $m\widehat{SL} = 75^\circ$, find $m\widehat{DS}$.
 $m\widehat{DS} = 95^\circ$



- Equilateral triangle CKV is inscribed in Circle A . Circle A consists of three equal inscribed angles. Determine the value of the inscribed angles.
 $m\angle C = m\angle V = m\angle K = 60^\circ$



This activity is designed to give students independent practice on what they have learned in this lesson. Student performance on this activity can provide valuable insight into which parts students may be struggling with.



Use these problems as an opportunity to work with small groups or individual students who are still struggling with the concept.

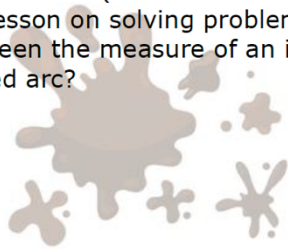
10.6.5 Wrap-Up: Reflection

Component Overview: Students will identify the most difficult or confusing point in this lesson.



10.6.5 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on solving problems using the relationship between the measure of an inscribed angle and its intercepted arc?



Answers will vary. The muddiest point in today's lesson was remembering whether it was half the value or the same value. The muddiest point in today's lesson was remembering the definition of an inscribed angle.

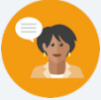


Consider pairing students to discuss their reflections to see if peers can clear up any misconceptions of the lesson.

10.7 Solving Problems Involving Intersecting Chords

Lesson Introduction

 Benchmark	MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.		 Learning Target	I can solve problems involving chords of a circle and their related angles.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- How can I use what I know about chords of a circle to solve problems? - What is the relationship between intersecting chords and the angles in the intersecting chords?			
 Lesson Narrative	In this lesson, students learn about angles in intersecting chords and how the relationships that exist in each can be used to solve problems. Students begin by exploring intersecting chords to discover the angle relationships that exist in intersecting chords. They will use this information to solve various types of problems throughout the lesson.			
 Component Summary	10.7.1 Warm-Up (~5 min.)	Evaluate the size of sectors drawn in a circle (<i>SE p. 154</i>)	10.7.4 Guided Practice (5-10 min.)	Solve problems involving angles in intersecting chords (<i>SE p. 157</i>)
	10.7.2 Exploration (5-10 min.)	Examine chords of a circle (<i>SE p. 154</i>)	10.7.5 Your Turn! (5 min.)	Solve a problem involving angles in intersecting chords (<i>SE p. 158</i>)
	10.7.3 Collaboration (10-15 min.)	Discover the relationship between the measures of angles of intersecting chords and their intersected arcs (<i>SE p. 155</i>)	10.7.6 Wrap-Up (~5 min.)	Reflect on the level of understanding of the lesson (<i>SE p. 159</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Chord - Angles of intersecting chords theorem <u>Additional Glossary:</u> - Vertical angles		- Protractor (Optional) Computers for GeoGebra (Optional) Graphing utility	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not recall the relationship of vertical angles. - Students may inaccurately label a perpendicular bisector when it doesn't meet the conditions of one.	Consider adding a simple problem with vertical angles as a quick refresher, so that it is at the front of their mind prior to 10.7.3 Collaboration.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder During the 10.7.1 Warm-Up, students are presented with a circle graph about student hours by subject in school. Students will be asked to share what they notice about the circle graph to provide a low entry point for all students.	MA.K12.MTR1.1: Participate in Effortful Learning The primary set of activities in this lesson are Collaboration and Exploration activities. Throughout these activities, students will build perseverance by modifying their methods, as needed, while solving a challenging task. They will work to help and support each other when attempting each problem.
 Differentiate Instruction	For 10.7.2 Exploration, students may need additional scaffolding to see the relationship between the angles and the intercepted arcs. Consider the use of sentence stems to build toward this discovery. Consider allowing students the opportunity to use GeoGebra as a resource for investigating angle measures of intersecting chords. Additional differentiation is notated throughout the lesson guide.	
Lesson Notes:		

10.7.1 Warm-Up: Sectors

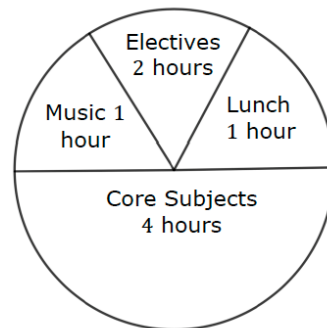
Component Overview: Students will answer three questions about sectors from a provided circle graph, in context.



10.7.1 Warm-Up: Sectors

1. What do you notice about the circle graph?

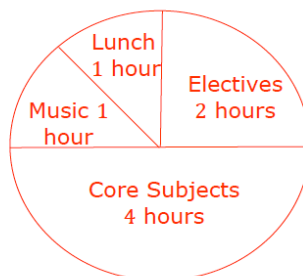
Answers will vary. I noticed that the number of hours in each sector aren't accurately represented by the sizes of each sector.



2. Identify a sector that is incorrectly sized.

Answers will vary. Music represents one hour, however, Electives represents two hours, but they are both the same sector size.

3. Redraw the circle graph so that the sizes are proportional to the number of hours.



For this activity, students are provided with a low-entry question first. This allows all students to actively engage in the content, regardless of their background. Provide some individual think time, and then allow students to share their thoughts with a partner before a whole-group review of the answers.

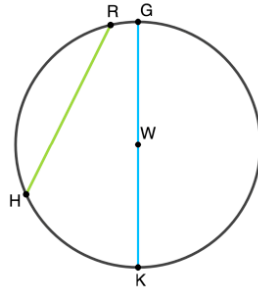
10.7.2 Exploration: Chords

Component Overview: Students will explore the concept of chords using a single example with three scaffolded questions.



10.7.2 Exploration: Chords

Circle W is shown where points R , G , K , and H lie on the circle.



1. What similarities do you notice between line segment RH and line segment GK ?

Answers will vary. Both line segments have points on the circle, both line segments are in the circle.

2. What differences do you notice between line segment RH and line segment GK ?

Answers will vary. Line segment GK goes through the center, line segment GK is longer.

A **chord** is a line segment that connects two points on the curve of a circle.

3. Complete the statements.

Line segment RH is
of the circle.

- ☐ a diameter
- ☒ a chord
- ☐ both a diameter and a chord

Line segment GK is
of the circle.

- ☐ a diameter
- ☐ a chord
- ☒ both a diameter and a chord



Consider making a list of students' answers to questions 1 and 2 for all to see. Students benefit from hearing the thinking of their peers.



- *What is the relationship between a diameter and a chord?*
- *How can a diameter also be classified as a chord?*

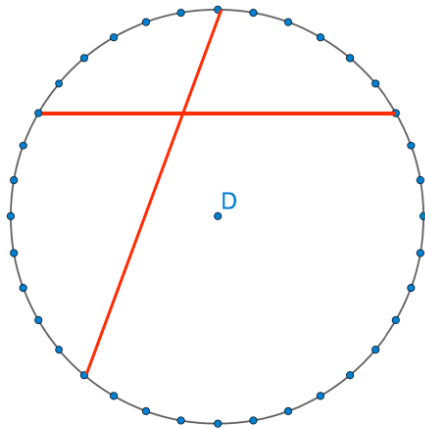
10.7.3 Collaboration: Angles in Intersecting Chords

Component Overview: Students will learn about the angles that exist when chords intersect. For this activity, students should work with a partner.



10.7.3 Collaboration: Angles in Intersecting Chords

1. With your partner, complete the following using Circle D.
 - a. Draw and label two chords that intersect inside the circle, making sure the endpoints fall on the points. When drawing the chords, do not draw the same chords as your partner and avoid drawing diameters.



- b. Choose one of the angles to measure and label. Then, complete the table.

	Angle Measure	Measure of Arc it Intercepts	Measure of the Arc the Vertical Angle Intercepts
Your Angle	120°	70°	170°
Your Partner's Angle	60°	30°	90°



For this activity, students will need a partner. Try and pair students together based on their performance throughout Unit 10. This can allow more students to get assistance during the facilitation of this activity, as well as learn with their peers.



If students struggle finding the measure of the arcs, consider asking the following:

- What is the degree measure of a whole circle?
- What do you notice about the dots on the circle? How many are there?
- What is the degree measure between each dot on the circle?



For question 1 part b, students will need to use their pre-requisite knowledge of vertical angles. Watch for students who may get this incorrect to determine if a refresher on vertical angles may be needed. Use this opportunity to spiral back to the relationship between the measures of vertical angles. Make connections that an angle and its vertical angle will have the same relationship with the intercepted arcs.

10.7.3 Collaboration: Angles in Intersecting Chords (continued)

c. Discuss the following with your partner.

- What do you notice about the values in the table?
- What do you think the relationship is between the angle measure and the two intercepted arcs?

Answers will vary. Our angle measure falls exactly halfway between the values of the two arc measures. An angle made by two intersecting chords is the average of the measures of the two arcs intercepted by the chords.

d. Write a formula that would generalize your discovery from the discussion you had in Part C.

$$m\angle A = \frac{m(\text{arc } 1) + m(\text{arc } 2)}{2}$$

e. Complete the statement.



Angles of Intersecting Chords Theorem

The angle formed inside of a circle by two intersecting chords is
☐ double
☐ equal
☒ half
 the sum of the measures of the arcs intercepted by the angle and its vertical angle.



Listen for student conversations when circulating the room. Ensure that the conversation for question 1 part c stays centered on the questions that go along with it, and not general noticing. The intent is for what they notice to serve as the springboard for their formula in question 1 part d. If needed, consider offering scaffolding support in the form of sentence stems:

- The sum of the measure of the two arcs an angle and its _____ intercepts is _____.
- The relationship between the _____ of the two arcs and the angle measure is _____.



For students that finish early, have them make connections between the discovery here to the measure of a central angle and its intercepted arc.

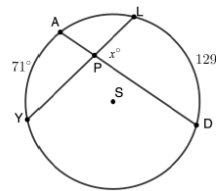
10.7.4 Guided Practice: Solving Problems Involving Angles in Intersecting Chords

Component Overview: Students will complete three practice problems on solving real-world and mathematical problems involving intersecting chords.

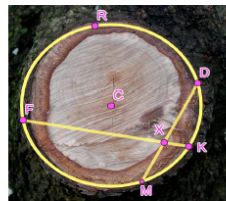


10.7.4 Guided Practice: Solving Problems Involving Angles in Intersecting Chords

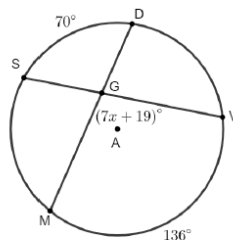
- Circle S is shown, where chords $\overline{AD}$ and $\overline{YL}$ intersect at point P . Find the value of x for $\angle LPD$, given $m\widehat{AY} = 71^\circ$ and $m\widehat{LD} = 129^\circ$.
 $x = 100$



- A tree is cut down and a circular tree stump is left. The top of the tree stump has two cracks that represent two chords, $\overline{FK}$ and $\overline{DM}$, intersecting at point X in Circle C . If $m\widehat{FRD} = 175^\circ$ and $m\widehat{KM} = 39^\circ$, what is the measure of $\angle FXD$?
 $m\angle FXD = 107^\circ$



- Circle A is shown, where chords $\overline{SV}$ and $\overline{MD}$ intersect at point G . Find the value of x , given $m\widehat{SD} = 70^\circ$, $m\widehat{MV} = 136^\circ$, and $m\angle MGV = (7x + 19)^\circ$.
 $x = 12$



For this activity, consider using these three questions as a gradual release practice:

For question 1, elicit information from students about what they notice in the circle. Use that information to connect students to the correct markings, and then model the correct thinking for solving.

For question 2, release students to try it on their own and then check with a partner before reviewing together as a class. Struggling readers may need additional processing time.

For question 3, fully release students to try this one on their own from start to finish before bringing the class back together to review the answer.



If students finish early, challenge them to name one piece of information that could be included that would allow them to find the measure of $\widehat{SM}$ and $\widehat{DV}$.

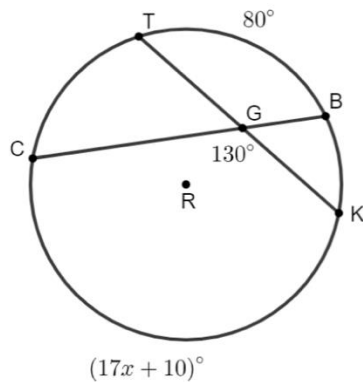
10.7.5 Your Turn!

Component Overview: Students will solve one mathematical problem using angles of intersecting chord relationships.



10.7.5 Your Turn!

1. Circle R is shown, where chords $\overline{TK}$ and $\overline{CB}$ intersect at point G . Find the value of x , given $m\widehat{TB} = 80^\circ$, $m\widehat{CK} = (17x + 10)^\circ$, and $m\angle CGK = 130^\circ$.



$$x = 10$$



For this question, students should complete it independently. This will provide valuable data to determine whether students are grasping the content from this lesson, and where they may be getting stuck.



For students who finish early, consider having them determine $m\widehat{CK}$.



10.7.6 Wrap-Up: Reflection

Answers will vary.

- | | | |
|----------------------|------------------------|-----------------------|
| I need help. | I'm getting there, but | I understand this, |
| I can't get started. | I need more practice. | and I feel confident. |

- | | | |
|----------------------|------------------------|-----------------------|
| I need help. | I'm getting there, but | I understand this, |
| I can't get started. | I need more practice. | and I feel confident. |

- | | | |
|----------------------|------------------------|-----------------------|
| I need help. | I'm getting there, but | I understand this, |
| I can't get started. | I need more practice. | and I feel confident. |



The information provided from the student here, coupled with what was observed during the lesson, can provide information for differentiating instruction in later lessons.

10.8 Solving Problems Involving Intersection of Secants

Lesson Introduction

 Benchmark	MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.			 Learning Target	I can solve problems involving secants drawn from a point external to the circle and their related angles.
 Benchmark Coherence	Building On			Working Toward	
	N/A			N/A	
 Essential Questions	What is the relationship between the angle formed by two secants intersecting in the exterior of a circle and intercepted arcs?				
 Lesson Narrative	In this lesson, students will learn about the angles formed by secants. Students will begin by exploring secants that intersect in the exterior of a circle to discover the exterior secant angle theorem. They will then apply this relationship to solve problems that involve algebraic expressions.				
 Component Summary	10.8.1 Warm-Up (~5 min.)	Reflecting on the importance of the ability to learn (<i>SE p. 160</i>)	10.8.3 Guided Practice (10-15 min.)	Solving problems involving angles formed by intersecting secants (<i>SE p. 161</i>)	
	10.8.2 Exploration (10 min.)	Discovering the relationship of angles formed by intersecting secants and their intercepted arcs (<i>SE p. 160</i>)	10.8.4 Your Turn! (15 min.)	Solving problems involving angles formed by intersecting secants (<i>SE p. 162</i>)	
	10.8.5 Wrap-Up (~5 min.)	Summarizing the lesson in a note to a classmate (<i>SE p. 163</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Secant - Exterior Secant Angle Theorem Additional Glossary: N/A		(Optional) GeoGebra Activity		Warm-Up 10.8.1 contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may reverse the subtraction of the arcs. - Students may apply central angle or inscribed angle relationships to secant angles.	Be careful not to give anything away too early during the 10.8.2 Exploration, as it is designed to be a discovery of the Exterior Secant Angle Theorem.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share During 10.8.2 Exploration, students will have independent time to think through and complete one section of the activity (a-c), and then collaborate with their partner in question 1 part d to compare conclusions before reviewing the formal definition of the exterior secant angle theorem.	MA.K12.MTR.5.1: Patterns and Structure For 10.8.3 Guided Practice and 10.8.4 Your Turn!, students will be applying what they learned in the prior activity about angles formed by secants to solve mathematical problems that involve algebraic expressions. To do this, students will need to focus on the relevant details within the problem, in order to correctly find the difference between the intercepted arcs.
 Differentiate Instruction	Some learners may benefit from seeing how angles are formed by secants to better understand this conceptually. To do this, consider using digital applets, like GeoGebra’s “Angles formed by Secants” (outside).	
Lesson Notes:		

10.8.1 Warm-Up: Satellite Operations Engineer

Component Overview: Play the Career Profile video on the Satellite Operations Engineer for the entire class. After students watch the video (3:35 minutes), prompt them to reflect on the importance of being able to be trained by answering the one question independently.



10.8.1 Warm-Up: Satellite Operations Engineer

1. Veronica Navarro is a satellite operations engineer. Before beginning her career, she received a bachelor's degree. One piece of advice she would give to high school students is to take as many classes in high school as possible because they are free. In her opinion, a degree beyond high school is needed in many careers because employers want to know a person can be trained and is capable of learning. Veronica's employer paid for her to receive additional degrees as a means to further develop her abilities and deepen her learning.

Why do you think it is important for companies to know you are capable of being trained?

Answers will vary. Companies want to hire employees that will continue to grow as they work. They will continuously improve their abilities to perform the duties of their position and their contributions to the company will grow more substantial over time.



Allow students to share their responses. Students benefit when they hear the thinking of their peers.

10.8.2 Exploration: Angles Formed by Secants

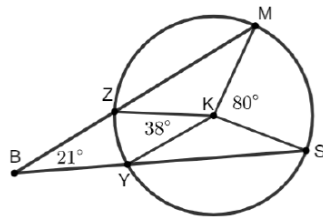
Component Overview: Students will explore angles formed by secants in the exterior of a circle through a four-part question series. For this activity, students will need a partner.



10.8.2 Exploration: Angles Formed by Secants

A **secant** is a line that passes through two points on a circle. In the interior of the circle, a secant line forms a chord.

1. $\angle MBS$ intersects Circle K , creating two intersecting arcs, where points M , Z , Y , and S are on the circle. The $m\angle ZKY = 38^\circ$, $m\angle SKM = 80^\circ$, and $m\angle MBS = 21^\circ$.



- a. Fill in the table.

Arc	Central Angle Measure	Arc Measure
$\widehat{MS}$	80°	80°
$\widehat{ZY}$	38°	38°

- b. Add and subtract the two arc measures.

$$\widehat{MS} + \widehat{ZY} = 118^\circ$$

$$\widehat{MS} - \widehat{ZY} = 42^\circ$$



This activity is designed to be an Exploration activity where students discover the Exterior Angle Theorem. Be careful not to give anything away too early in the activity when facilitating.



Watch student answers to question 1 part a. This can be an indicator of retention of the concepts from prior lessons. If students are stumped, refer them back to their notes from the central angle lesson earlier in the unit.

10.8.2 Exploration: Angles Formed by Secants (continued)

- c. Do the operations in Part B compare to the measure of $\angle MBS$? If so, explain.

The value of $m\angle MBS$ is half the difference of $\widehat{MS}$ and $\widehat{ZY}$.

- d. With your partner, write a formula that can be used to find the measure of an exterior angle of two secant lines.

$$m\angle MBS = \frac{1}{2}(\widehat{MS} - \widehat{ZY})$$



Exterior Secant Angle Theorem

The measure of an angle formed by two secants intersecting in the exterior of a circle is one-half the difference of the measures of the intercepted arcs.



Some learners may benefit from seeing how angles are formed by secants to better understand this conceptually. To do this, consider digital applets, like GeoGebra's "Angles formed by Secants" (outside).

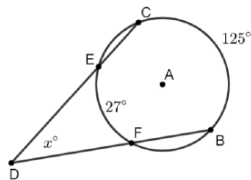
10.8.3 Guided Practice: Solving Problems Involving Angles Formed by Secants

Component Overview: Students will complete two structured practice questions where they will apply the relationship between angles formed by secants and their intercepted arcs to solve algebraic problems.



10.8.3 Guided Practice: Solving Problems Involving Angles Formed by Secants

- Circle A has points C , E , F , and B on the circle. Secant lines $\overline{CD}$ and $\overline{BD}$ intersect at point D outside the circle. The $m\widehat{CB} = 125^\circ$, $m\widehat{EF} = 27^\circ$, and $m\angle CDB = x^\circ$. What is the value of x ?



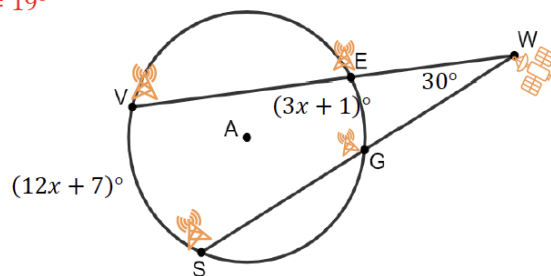
$$x = 49$$

- A satellite, point W , sends signals $\overline{VW}$ and $\overline{SW}$ to Earth. Signal $\overline{VW}$ hits tower E before reaching tower V . Similarly, signal $\overline{SW}$ hits tower G before reaching tower S . Scientists are interested in finding the measure of the intersecting arcs created by the two signals. They know $m\angle W = 30^\circ$, $m\widehat{VS} = (12x + 7)^\circ$, and $m\widehat{EG} = (3x + 1)^\circ$.

What is $m\widehat{VS}$ and $m\widehat{EG}$?

$$m\widehat{VS} = 79^\circ$$

$$m\widehat{EG} = 19^\circ$$



For this activity, consider beginning with a teacher-modeled activity for question 1. Students can benefit from hearing structured thinking out loud on determining what is known from the problem, what relationships can be determined, and how those relationships can be used to solve problems.



Question 2 provides a great call back to the career explored during the 10.8.1 Warm-Up. Consider highlighting this with the students to help debunk the myth that this is not applicable in the real-world.

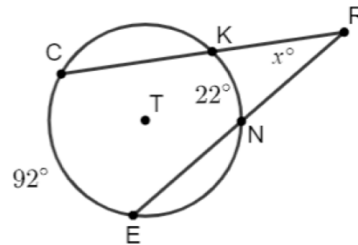
10.8.4 Your Turn!

Component Overview: Students will complete three practice questions where they will use their knowledge of the angle relationships between angles formed by secants and their intercepted arcs to solve algebraic problems.



10.8.4 Your Turn!

1. Circle T has points C , K , N , and E on the circle. Secant lines $\overline{CR}$ and $\overline{ER}$ intersect at point R outside the circle. The $m\widehat{CE} = 92^\circ$, $m\widehat{KN} = 22^\circ$, and $m\angle CRE = x^\circ$. What is the value of x ?

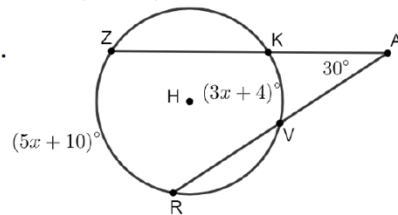


$$x = 35$$

2. Circle H has points Z , K , V , and R on the circle. Secant lines $\overline{ZA}$ and $\overline{RA}$ intersect at Point A outside the circle. The $m\widehat{ZR} = (5x + 10)^\circ$, $m\widehat{KV} = (3x + 4)^\circ$, and $m\angle ZAR = 30^\circ$.

Find the measure of $\widehat{ZR}$.

$$m\widehat{ZR} = 145^\circ$$



This activity is designed to give students independent practice on what they have learned in this lesson. Student performance on this activity can provide valuable insight into which parts students may be struggling with.



For students who finish early, have them create their own problem involving the relationship between intersecting secants and their intercepted arcs. Set the requirement that the measures of arcs and the angle must be written as algebraic expressions. Students should be cautious in creating a problem that, appropriately, has the closer arc measure smaller than the measure of the arc that is further from the angle.

10.8.5 Wrap-Up: Cool Down

Component Overview: Students will write a short note to a friend who missed class. This note will provide their own summary of what they learned today.



10.8.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed class summarizing today's lesson.

Answers will vary. Today we solved problems involving secants drawn from a point external to the circle. There were several real-world problems to help us figure out how to apply the concepts.



For students who might struggle representing their understanding, consider alternative ways to assess or measure learning for this lesson.

- If technology is available, students could record themselves explaining their summary.
- Students could provide detailed written methods of finding angles of intersecting chords.

10.9 Solving Problems Involving Tangents

Lesson Introduction

 Benchmark	MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.		 Learning Target	I can solve problems related to tangents and secants and the measures of their intercepted arcs.
 Benchmark Coherence	Building On		Working Toward	
	N/A	N/A		
 Essential Questions	- What is the relationship between a line tangent to a circle and the radius of that circle? - What is the relationship between the angles created by two secants, two tangents, or one secant and one tangent intersecting at an external point on the circle?			
 Lesson Narrative	Throughout this lesson, students are introduced to tangent lines and the angles formed by intersecting tangents and secants. Students will begin by exploring tangent lines and the relationships with the circle the line runs tangent to. They will then collaborate to discover angles formed by intersecting tangents and secants. Students will use this knowledge to solve mathematical problems for missing arc and angle measures.			
 Component Summary	10.9.1 Warm-Up (~5 min.)	Decoding a number puzzle (<i>SE p. 164</i>)	10.9.3 Collaboration (10-15 min.)	Discovering the relationship between angles formed by intersecting tangents or tangents and secants and their intercepted arcs (<i>SE p. 166</i>)
	10.9.2 Exploration (5-10 min.)	Observing relationships of lines drawn from the center of a circle to a line tangent to the circle (<i>SE p. 165</i>)	10.9.4 Your Turn! (10-15 min.)	Solving problems involving the intersection of tangents or a tangent and secant (<i>SE p. 168</i>)
	10.9.5 Wrap-Up (~5 min.)	Reflecting on what is not known yet and what is understood (<i>SE p. 169</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Tangent - External Angle to a Circle <u>Additional Glossary:</u> - N/A		(Optional) GeoGebra Activities	N/A

 Teacher Tips	Common Misconceptions • Students may confuse a secant line with a tangent line and vice-versa. • Students may think the angle relationships are the same as one of the other angle relationships in circles. • Students may subtract the wrong intercepted arcs.	Things to Consider • Be cautious not to give away anything in 10.9.2 Exploration, as it is designed to be a discovery activity for students.
	Literacy and Language Routines Think-Pair-Share 10.9.3 Collaboration provides an opportunity to leverage the Think-Pair-Share routine. In question 1, students should think through the relationship formed before discussing with a partner for question 1 part c, as well as for question 2 part b. Students should be prepared to share their thoughts during the whole-class discussion.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.1.1: Participate in Effortful Learning In 10.9.3 Collaboration, students will work together with partners to work with angles formed by tangents and secants. During the process, students will be encouraged to stay engaged and maintain a positive mindset when working to solve tasks.
	 Differentiate Instruction Consider using additional visuals, such as GeoGebra’s “An Angle Formed by Two Tangent Lines and Major and Minor Arcs” and “Angle Formed by A Secant and A Tangent,” to help students better conceptualize the concept of a tangent and its relationship to the radius in the circle for visual and kinesthetic elements. Students will have the chance to draw their own conclusions and discuss their rationale with peers and teacher, providing an opportunity for auditory learners to verbalize their conception.	
Lesson Notes:		

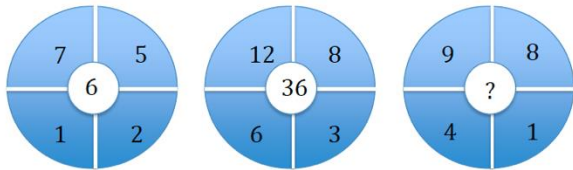
10.9.1 Warm-Up: Number Puzzle

Component Overview: Students will find the rule to determine the missing number following a number pattern.



10.9.1 Warm-Up: Number Puzzle

- In the circles, the outer numbers are used to calculate the number in the middle. Find the rule to determine what number should replace the question mark in the third circle.



The number in the middle of each circle can be calculated by multiplying the difference of the two top numbers and the sum of the two bottom numbers: $(7 - 5)(1 + 2) = 6$, $(12 - 8)(6 + 3) = 36$. The missing number is $(9 - 8)(4 + 1) = 5$.



Consider leveraging a Think-Pair-Share routine for this activity. Allow students some individual think time, and then have them work with a partner to discuss their work and rationale. Engage students in a whole-group discussion around their methods for determining the pattern.

10.9.2 Exploration: Intersecting Tangent Lines

Component Overview: Students will explore the concept of tangent lines. For this activity, students will need a partner.

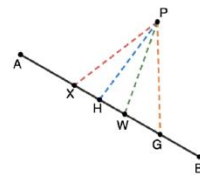


10.9.2 Exploration: Intersecting Tangent Lines

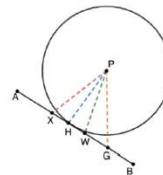
1. With your partner, answer the following.

- a. Without measuring, which dashed line represents the shortest distance from point P to line $\overline{AB}$? Explain your answer.

Answer will vary. $\overline{HP}$ represents the shortest distance from P to line AB



- b. Point P is the center of a circle that intersects $\overline{AB}$. In the table, describe your observations of each dashed segment in relation to the center of Circle P .



Segment	Relationship with the Center of Circle P
$\overline{PG}$	<i>Greatest distance to the circle center</i>
$\overline{PW}$	<i>Second-shortest distance to circle center</i>
$\overline{PH}$	<i>Shortest distance to the circle center</i>
$\overline{PX}$	<i>Second-greatest distance to circle center</i>

- c. Discuss with your partner the relationship the blue segment has with the center of Circle P .
The blue line looks a lot like it could be the radius of the circle.

A **tangent** is a line or segment that intersects a circle at exactly one point.



If a line is tangent to a circle, then it is perpendicular to a radius, which ends at the point of tangency.



What type of line represents the shortest distance between a point and a line?



Consider using a sentence stem for students who struggle answering question 1 part c:

- If point H lies on the circle, then the blue line looks like a _____ and will be the _____ distance to $\overline{AB}$.

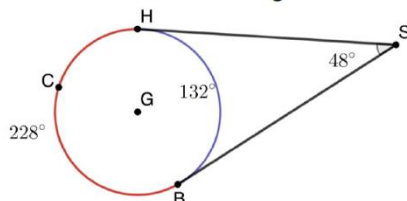
10.9.3 Collaboration: Angles Formed by Tangents and Secants

Component Overview: Students will collaborate to work with the angles formed by tangents and secants and their relationship to intercepted arcs. For this activity, students will need a partner.



10.9.3 Collaboration: Angles Formed by Tangents and Secants

1. Circle G with tangents $\overline{SH}$ and $\overline{SB}$ is shown. Use the diagram to answer the following.



- a. Complete the table.

	Measure
Major Arc HCB	228°
Minor Arc HB	132°
$\angle HSB$	48°

- b. Use the values for the major and minor arcs of Circle G to determine a relationship between the arcs and the measure of the exterior angle formed by the two tangents. The relationship can be described using words or equations.
The measure of the exterior angle is half the difference between the measures of the two intercepted arcs.
- c. Ask two other partner pairs to share their relationships. Record their relationships and ask them to initial verifying you have written it correctly. If their description or equation is different than yours, discuss the differences and determine if your description or equation needs to be revised.

Description or Equation	Initials
Answers will vary.	



For a visual element, consider applets, like GeoGebra's "An Angle Formed by Two Tangent Lines and Major and Minor Arcs," for students who struggle seeing the relationship between the angle and the intercepted arc.



Consider asking the following as an opportunity to spiral back to the definitions of major and minor arcs:

- What is a minor arc? How do you name a minor arc?
- What is a major arc? How do you name a major arc?

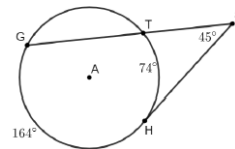
10.9.3 Collaboration: Angles Formed by Tangents and Secants (continued)

- d. Explain how to find the measure of an angle formed by two tangents drawn to a circle from a point outside that circle. Be sure to use the term(s) "intercepted arc(s)" at least once in the description.

Answers will vary. The measure of an angle formed by two tangents drawn to a circle from a point outside that circle is equal to half the difference of the measures of the intercepted arcs.

2. Circle A shows $\angle GSH$ formed by the intersection of tangent $\overline{SH}$ and secant $\overline{GS}$ outside the circle at point S . $m\widehat{GH} = 164^\circ$, $m\widehat{TH} = 74^\circ$, and $m\angle GSH = 45^\circ$.

- a. Use the values for $\widehat{GH}$ and $\widehat{TH}$ of Circle A to determine the relationship between the arcs and the measure of the exterior angle formed by the tangent and the secant.



$$\frac{164 - 74}{2} = \frac{90}{2} = 45^\circ$$

- b. Discuss the relationship with your partner.

Students do not need to write an explanation here, but a sample response is provided for reference. It does not matter if the lines are both tangent, secant, or a combination of the two, the angle will always be equal to the difference of the two intercepting arcs divided by 2.



The formula for the measure of an **External Angle to a Circle** formed by two tangents, two secants, or a tangent and a secant is

$$\text{Measure of angle formed} = \frac{1}{2}(\text{difference of intercepted arcs})$$

OR

$$\text{Measure of angle formed} = \frac{1}{2}(\text{larger arc} - \text{smaller arc})$$



For a visual element, consider applets, like GeoGebra's "Angle Formed by a Secant and a Tangent," for students who struggle seeing the relationship between the angle and the intercepted arc.

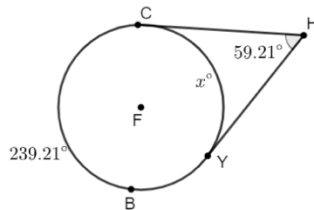
10.9.4 Your Turn!

Component Overview: Students will complete two practice problems independently on finding the missing angle measures using their knowledge of angles formed by tangent and secant lines.



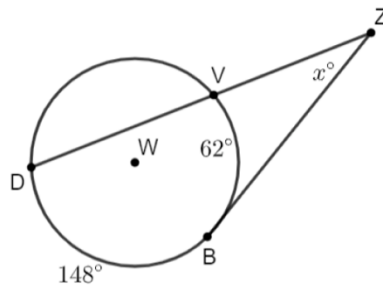
10.9.4 Your Turn!

- Circle F has points C and Y on the circle. Tangents $\overline{CH}$ and $\overline{YH}$ intersect at point H outside the circle. $m\widehat{CBY} = 239.21^\circ$, $m\widehat{CY} = x^\circ$, and $m\angle CHY = 59.21^\circ$. Find the value of x .



$$x = 120.79$$

- Circle W has points D , V , and B on the circle. Tangent $\overline{BZ}$ and secant $\overline{DZ}$ intersect at Point Z outside the circle. $m\widehat{DB} = 148^\circ$, $m\widehat{VB} = 62^\circ$, and $m\angle DZB = x^\circ$. What is $m\angle DZB$?



$$x = 43$$



For students who struggled in the prior lesson, use the time in this activity to work with individual students or small groups of students who may need additional support. Consider using question 1 as a “we do” style, and release students you worked with in small groups to try question 2 on their own.

10.9.5 Wrap-Up: Reflection

Component Overview: Students will complete a reflection of what they do not understand yet and what they can explain.



10.9.5 Wrap-Up: Reflection

Think back over this unit. If necessary, look back over the different lessons. Then, complete the sentences by checking all that apply.

Answers will vary.

1. I don't understand

- ☐ arc length
- ☐ area of a sector
- ☐ converting between radians and degrees
- ☐ measures of arcs and related angles

...YET.



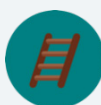
Consider providing students with the opportunity to expand on things they do not understand by having students write out questions they still have on Unit 10.

2. I can explain

- ☐ how to solve problems involving arc length.
- ☐ how to solve problems involving area of a sector.
- ☐ converting angle measures between degrees and radians.
- ☐ how to solve problems involving: measures of arcs and related angles including central, inscribed, and intersections of a chords, secants, and tangents.

10.H1 Constructing Tangent Lines

Lesson Introduction

 Benchmark	MA.912.GR.5.5 Given a point outside a circle, construct a line tangent to the circle that passes through the given point.		 Learning Target	I can construct a line tangent to a circle that passes through a given point outside the circle.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	How do you construct a line tangent to a circle that passes through a given point outside the circle?			
 Lesson Narrative	Students will learn how to construct a line tangent to a circle that passes through a given point outside the circle with a compass and a ruler. Students will begin by collaborating to learn about tangent lines of circles and the steps to constructing tangent lines.			
 Component Summary	10.H.1 Warm-Up (~5 min.)	Determining radio signal distance and the signal's relationship to the Earth (<i>SE p. 170</i>)	10.H.3 Your Turn! (10-15 min.)	Constructing a tangent line and examining examples of constructions (<i>SE p. 172</i>)
	10.H.2 Collaboration (20-25 min.)	Constructing a tangent to a circle (<i>SE p. 170</i>)	10.H.4 Wrap-Up (~5 min.)	Constructing a tangent to a circle (<i>SE p. 173</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> Tangent Line		- Compass - Ruler (Optional) Digital Version of Construction	
			Additional Preparation	
			This lesson is designated for students enrolled in the honors version of Geometry only.	

 Teacher Tips	Common Misconceptions • Students may skip a step in the construction process. • Students may think any line that intersects a circle in one or more points is a tangent.	Things to Consider • Consider a refresher on the use of the tools and steps as part of the launch of the lesson.
	Literacy and Language Routines Think-Pair-Share 10.H.2 Collaboration provides students a collaborative opportunity to work together with constructing tangent lines. Students will have the opportunity for individual thinking and practice before collaborating with a partner to share strategies and results.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.1.1: Participate in Effortful Learning 10.H.2 Collaboration provides students an opportunity to stay engaged through an interactive lesson with constructions.
	 Differentiate Instruction Consider showing students a digital version of the step-by-step construction process. This will allow entry points for visual and kinesthetic learners. Engage the class in a discussion about the steps taken in the construction to allow for an auditory entry point.	
Lesson Notes:		

10.H1.1 Warm-Up: Radio Signals

Component Overview: Students will explore the distance a radio signal would go based on their height.



10.H1.1 Warm-Up: Radio Signals

- Two-way radio communications help climbers stay in touch with base camp and contact the other expeditioners climbing Mt. Everest. The separate expeditions reserve a central channel where the various groups can share information on weather conditions and route changes or call for rescue.

The equation shows the maximum theoretical range of a radio signal, in miles, using a line-of-sight calculation, which considers the curvature of the Earth and presumes the Earth to be a perfect sphere – which of course it is not. This also assumes the radio signal is transmitted with sufficient power that maintains enough energy to travel the distance.

$$\text{Horizon(mi)} = 1.23 \times \sqrt{\text{height(ft)}}$$



- Use the formula to find the distance the radio signal would go based on your height.

Answers will vary.

$$5\text{ft } 6\text{in} = 5.5\text{ft} \quad 1.23 \times \sqrt{5.5} = 2.885 \text{ miles}$$

- Explain the relationship of the radio signal range (red line) and Earth.

Answers may vary.

The radio signal range is similar to a tangent line that touches the Earth at only one point.



Why would a lower height affect the path of the radio signal?

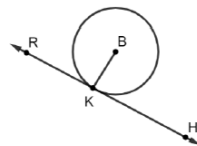
10.H1.2 Collaboration: Tangent Lines of Circles

Component Overview: Students will work together to construct tangent lines of circles. For this activity, students will need a compass and a ruler.



10.H1.2 Collaboration: Tangent Lines of Circles

1. Circle B is shown, with point K on the circle and $\overleftrightarrow{RK}$ tangent to circle B .



- a. Write two facts about circle B and tangent line $\overleftrightarrow{RK}$.
Answers will vary.

1. *Line RK intersects circle B at one point*
2. *Line RK is perpendicular to the radius BK*

- b. Share your facts with your partner and write down one additional fact.

3. *Answers will vary*
 $\angle BKH$ and $\angle BKR$ are right angles.

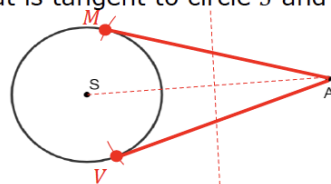
2. Complete the statement.

A line tangent to a circle touches a circle at

point(s), and the line is ☐ parallel ☒ perpendicular to the radius of the circle drawn to the point of tangency.

- ☐ one
- ☐ two
- ☐ three

3. To construct a line tangent to a circle, you need a compass and a ruler.
 - a. Work with your partner and use the steps below to construct a line that is tangent to circle S and goes through point A .



Consider pausing the activity after question 2 to bring the class together to ensure that everyone is on the same page with the correct understanding of the concept. A misconception here could cause errors later in the lesson.

10.H1.2 Collaboration: Tangent Lines of Circles (continued)

1.	Use the ruler to draw a line from the center, point S , to the given point .
2.	Construct the perpendicular bisector of $\overline{SA}$ to find its midpoint.
3.	Set the compass width to the distance from the midpoint of $\overline{SA}$ to the circle's center, point S .
4.	Keeping the compass at the midpoint, mark an arc across the circle at two places, creating M and V .
5.	Draw the two tangent lines from A to M and A to V .



To complete question 3, students will need to have a compass and a ruler available. Consider having these prepared and distributed prior to this question in the activity to maintain instructional momentum.

4. Jamie wrote the steps for constructing a line tangent to a circle on note cards, including the steps of the perpendicular bisector, but the cards got out of order.

- a. Work with your partner to order the steps 1 – 8.

6	Place the compass on the midpoint of the line segment and adjust its lengths to reach the center.
2	Place the compass at one end of the line segment and adjust the compass to slightly longer than half the line segment.
5	Place the ruler where the arcs above and below the line cross, and draw the perpendicular line.
7	Keeping the same width and compass point at the midpoint, draw a circle.
3	Draw arcs above and below the line.
8	Where the new circle crosses the original circle, create a point and draw the tangent line.
1	Draw a line connecting the center of the circle and the point outside the circle.
4	Keeping the same compass width, draw arcs above and below the line from the other end of the line segment.



Consider showing students the digital version of the step-by-step construction process for a visual element.

10.H1.2 Collaboration: Tangent Lines of Circles (continued)

- b. Explain how Jamie's method is different.

Answers will vary. Jamie drew an entire circle for step 7 and uses that circle for step 8, while the steps in problem 3 only draws 2 arcs that cross the circle.

- c. Explain why Jamie's method works.

Answers will vary. Jamie's method meets the same conditions as the method in problem 3.



Consider adding a layer of engagement by taking these steps in 1-8 and printing them onto colored paper and cutting them into strips. This will provide a kinesthetic element and allow students to sort the steps around and place them into the correct order.



At the conclusion of this activity, engage students in a discussion about their answers to question 4 parts b and c. It is important for students to understand the various methods and why they work.

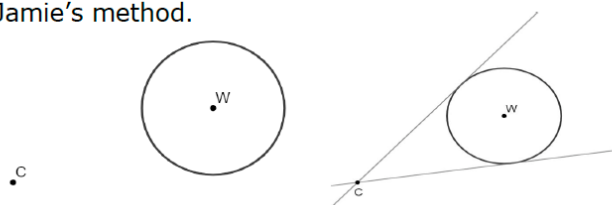
10.H1.3 Your Turn!

Component Overview: Students will independently complete two tangent construction practice problems.



10.H1.3 Your Turn!

- Given circle W and point C outside the circle, construct a line tangent to circle W going through point C using Jamie's method.



- Naomi and Xavier were practicing constructing a line tangent to circle B going through point A , but their results did not match.

Naomi's Construction	Xavier's Construction

- Who correctly drew a line tangent to circle B going through point A ?

Xavier

- Explain the mistake the other student made.

Answers will vary. Naomi's construction does not go through point A.



Consider using this time to work with individual students or small groups based on those who were struggling during the 10.H.2 Collaboration.



Have students who finish early swap their constructions from question 1. Students will then evaluate the classmate's work and offer suggestions for improvement or correction.

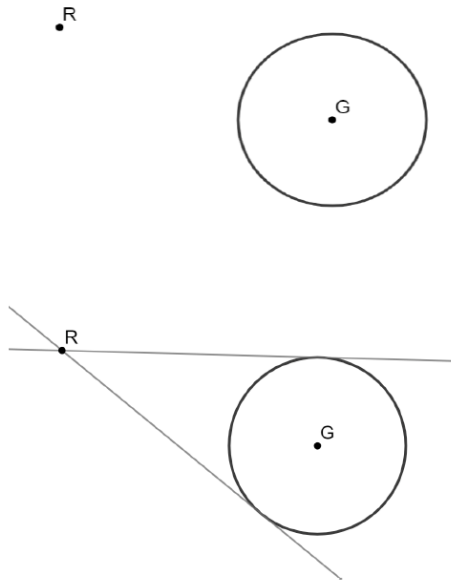
10.H1.4 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door with constructing a line tangent to a circle.



10.H1.4 Wrap-Up: Ticket Out the Door

1. Given circle G and point R outside the circle, construct a line tangent to circle G going through point R .



Students construction on this activity can provide valuable data as to whether students understand the content of this lesson.